```
 GET
  FILE='C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav'.
DATASET NAME DataSet1 WINDOW=FRONT.
* ===================================================================
* BI-DMQ Analysis v30 - Comprehensive SPSS Syntax
* Sample: n = 186 (full sample, no exclusions)
* Author: Hoda Rezvanjoo
* Purpose: Single-run, reproducible analysis for journal submission
* ===================================================================

* -------------------- 0. SETUP --------------------
* Adjust GET FILE path to where you saved the .sav file.

GET FILE='BI_DMQ_data_v30.sav'.

>Error # 61 in column 10.  Text: BI_DMQ_data_v30.sav
>The filename is not valid.
>Execution of this command stops.
DATASET NAME RawData WINDOW=FRONT.

* Variable labels for demographics.
VARIABLE LABELS
  Company_Size 'Company size band (1=<50, 2=50-99, 3=100-249)'
  Job_Position 'Job position (1=CEO, 2=Senior Manager, 3=Middle Manager, 4=Ope
rational Manager, 5=Other)'
  Industry 'Industry (1=Manufacturing, 2=Retail, 3=Service, 4=Technology, 5=Ot
her)'.

VALUE LABELS Company_Size 1 '<50 employees' 2 '50-99 employees' 3 '100-249 emp
loyees'.
VALUE LABELS Job_Position 1 'CEO' 2 'Senior Manager' 3 'Middle Manager' 4 'Ope
rational Manager' 5 'Other'.
VALUE LABELS Industry 1 'Manufacturing' 2 'Retail' 3 'Service' 4 'Technology'
5 'Other'.

EXECUTE.

* -------------------- 1. DATA QUALITY CHECKS --------------------

* 1a. Sample size and missing values.
```

```
DESCRIPTIVES VARIABLES=ALL /STATISTICS=MEAN STDDEV MIN MAX.
```

Descriptives

Notes

Output Created		18-JUN-2026 22:32:46
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=ALL /STATISTICS=MEAN STDDEV MIN MAX.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

```
[RawData] C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
```

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Respondent ID	186	1.00	186.00	93.5000	53.83772
Company size band (1=<50, 2=50-99, 3=100-249)	186	1.00	3.00	1.9355	.78232
Job position (1=CEO, 2=Senior Manager, 3=Middle Manager, 4=Operational Manager, 5=Other)	186	1.00	5.00	3.7742	1.29556
Industry (1=Manufacturing, 2=Retail, 3=Service, 4=Technology, 5=Other)	186	1.00	5.00	3.1398	1.23544
Strategic_BI_Availability	186	1.00	5.00	3.6452	.95459
Strategic_BI_Alternatives	186	1.00	5.00	3.7419	.91714
Strategic_BI_Alignment	186	1.00	5.00	3.8118	.87120
Tactical_BI_Availability	186	1.00	5.00	3.7097	.90142
Tactical_BI_Resource_Alloc	186	1.00	5.00	3.7688	.82874
Tactical_BI_Data_Decisions	186	1.00	5.00	3.9086	.83000
Operational_BI_Availability	186	1.00	5.00	3.7903	.95528
Operational_BI_Agility	186	1.00	5.00	3.8495	.85039
Operational_BI_Quality	186	1.00	5.00	3.8280	.86523
BI_Improves_Decisions	186	1.00	5.00	3.8495	.86301
BI_Speed_Accuracy	186	1.00	5.00	3.7742	.96567
BI_Data_Inform	186	1.00	5.00	3.9355	.87988
BI_Alternative_Solutions	186	1.00	5.00	3.7097	.89540
BI_Cost_Barrier	186	1.00	5.00	3.5215	1.04083
BI_Training_Limit	186	1.00	5.00	3.7634	.96308
BI_Flexibility_Issue	186	1.00	5.00	3.3280	.98364
BI_Data_Integration_Issue	186	1.00	5.00	3.4839	.90778
Valid N (listwise)	186				

```
* 1b. Frequency of demographics.
```

```
FREQUENCIES VARIABLES=Company_Size Job_Position Industry
  /STATISTICS=NONE.
```

Frequencies

Notes

Output Created		18-JUN-2026 22:32:46
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=Company_Size Job_Position Industry /STATISTICS=NONE.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Statistics

		Company size band (1=<50, 2=50-99, 3=100-249)	Job position (1=CEO, 2=Senior Manager, 3=Middle Manager, 4=Operational Manager, 5=Other)	Industry (1=Manufacturing, 2=Retail, 3=Service, 4=Technology, 5=Other)
N	Valid	186	186	186
	Missing	0	0	0

Frequency Table

Company size band (1=<50, 2=50-99, 3=100-249)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	<50 employees	63	33.9	33.9	33.9
	50-99 employees	72	38.7	38.7	72.6
	100-249 employees	51	27.4	27.4	100.0
	Total	186	100.0	100.0	

Job position (1=CEO, 2=Senior Manager, 3=Middle Manager, 4=Operational Manager, 5=Other)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	CEO	13	7.0	7.0	7.0
	Senior Manager	22	11.8	11.8	18.8
	Middle Manager	37	19.9	19.9	38.7
	Operational Manager	36	19.4	19.4	58.1
	Other	78	41.9	41.9	100.0
	Total	186	100.0	100.0	

Industry (1=Manufacturing, 2=Retail, 3=Service, 4=Technology, 5=Other)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Manufacturing	14	7.5	7.5	7.5
	Retail	51	27.4	27.4	34.9
	Service	53	28.5	28.5	63.4
	Technology	31	16.7	16.7	80.1
	Other	37	19.9	19.9	100.0
	Total	186	100.0	100.0	

```
* 1c. Identify near-straightline cases (zero variance across 13 BI/DMQ items).
COMPUTE BI_DMQ_SD = SD(Strategic_BI_Availability, Strategic_BI_Alternatives, S
trategic_BI_Alignment,
                     Tactical_BI_Availability, Tactical_BI_Resource_Alloc, Ta
ctical_BI_Data_Decisions,
```

```
                      Operational_BI_Availability, Operational_BI_Agility, Ope
rational_BI_Quality,
                      BI_Improves_Decisions, BI_Speed_Accuracy, BI_Data_Inform
, BI_Alternative_Solutions).
VARIABLE LABELS BI_DMQ_SD 'Within-respondent SD across 13 BI/DMQ items'.
EXECUTE.

FREQUENCIES VARIABLES=BI_DMQ_SD /STATISTICS=NONE /FORMAT=NOTABLE
  /HISTOGRAM.
```

Frequencies

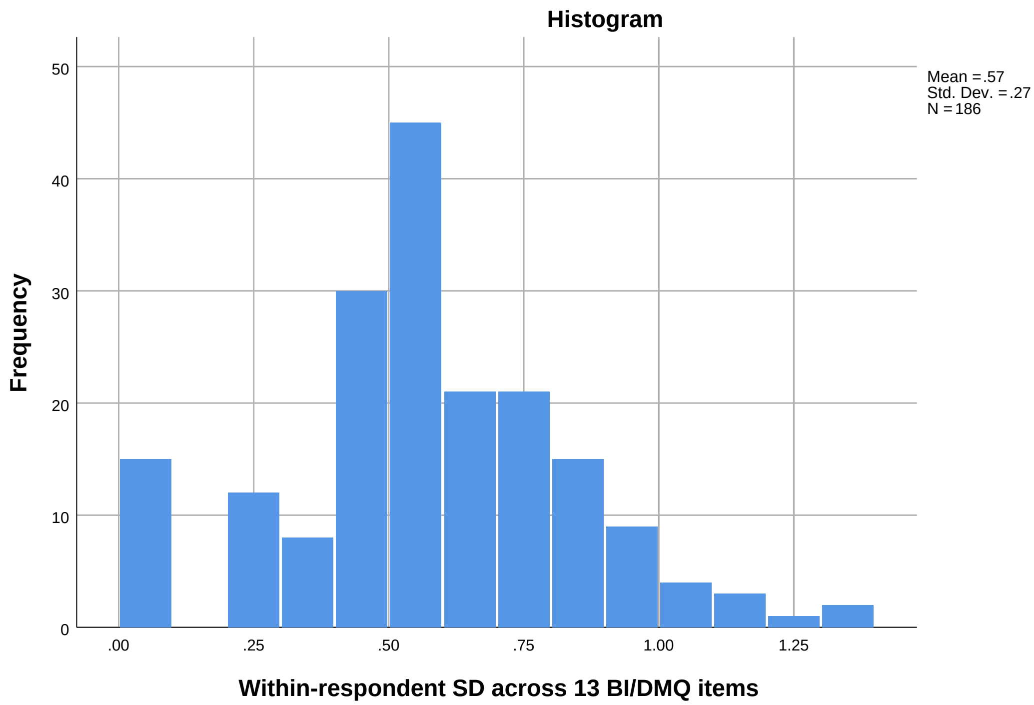
Notes

Output Created		18-JUN-2026 22:32:46
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data.
Syntax		FREQUENCIES VARIABLES=BI_DMQ_SD /STATISTICS=NONE /FORMAT=NOTABLE /HISTOGRAM.
Resources	Processor Time	00:00:00.67
	Elapsed Time	00:00:00.82

Statistics

Within-respondent SD across 13 BI/DMQ items

N	Valid	186
	Missing	0




```
 * List IDs with SD = 0 (perfect straight-liners).
TEMPORARY.
SELECT IF (BI_DMQ_SD = 0).
LIST VARIABLES=ID Company_Size Job_Position Industry.
```

List

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	15
Syntax		LIST VARIABLES=ID Company_Size Job_Position Industry.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

ID	Company_Size	Job_Position	Industry
55.00	3.00	5.00	4.00
57.00	2.00	3.00	5.00
58.00	2.00	4.00	2.00
59.00	2.00	3.00	5.00
119.00	2.00	5.00	2.00
120.00	2.00	4.00	2.00
121.00	2.00	4.00	2.00
122.00	2.00	3.00	2.00
173.00	2.00	4.00	4.00
175.00	2.00	2.00	1.00
176.00	2.00	4.00	4.00
177.00	2.00	4.00	3.00
178.00	3.00	3.00	4.00
181.00	2.00	2.00	3.00
182.00	2.00	2.00	5.00

Number of cases read: 15   Number of cases listed: 15

```
 * -------------------- 2. COMPUTE COMPOSITE MEANS --------------------

COMPUTE Strategic_Mean = MEAN(Strategic_BI_Availability, Strategic_BI_Alternat
ives, Strategic_BI_Alignment).
COMPUTE Tactical_Mean = MEAN(Tactical_BI_Availability, Tactical_BI_Resource_Al
loc, Tactical_BI_Data_Decisions).
COMPUTE Operational_Mean = MEAN(Operational_BI_Availability, Operational_BI_Ag
ility, Operational_BI_Quality).
COMPUTE DMQ_Mean = MEAN(BI_Improves_Decisions, BI_Speed_Accuracy, BI_Data_Info
rm, BI_Alternative_Solutions).
COMPUTE Challenges_Mean = MEAN(BI_Cost_Barrier, BI_Training_Limit, BI_Flexibil
ity_Issue, BI_Data_Integration_Issue).
COMPUTE Overall_BI_Mean = MEAN(Strategic_BI_Availability, Strategic_BI_Alterna
tives, Strategic_BI_Alignment,
                                Tactical_BI_Availability, Tactical_BI_Resource
```

```
_Alloc, Tactical_BI_Data_Decisions,
                                Operational_BI_Availability, Operational_BI_Ag
ility, Operational_BI_Quality).
COMPUTE Managerial_BI_Mean= MEAN(Strategic_BI_Availability, Strategic_BI_Alte
rnatives, Strategic_BI_Alignment,
                                   Tactical_BI_Availability, Tactical_BI_Resou
rce_Alloc, Tactical_BI_Data_Decisions).
EXECUTE.

VARIABLE LABELS
  Strategic_Mean 'Strategic BI utilisation composite'
  Tactical_Mean 'Tactical BI utilisation composite'
  Operational_Mean 'Operational BI utilisation composite'
  DMQ_Mean 'Decision-making quality composite'
  Challenges_Mean 'BI challenges composite'
  Overall_BI_Mean 'Overall BI utilisation (all 9 items)'
  Managerial_BI_Mean 'Managerial BI (Strategic + Tactical)'.

* -------------------- 3. DESCRIPTIVES OF COMPOSITES --------------------

DESCRIPTIVES VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean DMQ_Mean
                      Challenges_Mean Overall_BI_Mean Managerial_BI_Mean
  /STATISTICS=MEAN STDDEV MIN MAX SKEWNESS KURTOSIS.
```

Descriptives

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean DMQ_Mean Challenges_Mean Overall_BI_Mean Managerial_BI_Mean /STATISTICS=MEAN STDDEV MIN MAX SKEWNESS KURTOSIS.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation	Skewness
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic
Strategic BI utilisation composite	186	1.00	5.00	3.7330	.77329	-.725
Tactical BI utilisation composite	186	1.00	5.00	3.7957	.70610	-.750
Operational BI utilisation composite	186	1.33	5.00	3.8226	.74346	-.685
Decision-making quality composite	186	1.50	5.00	3.8172	.74538	-.642
BI challenges composite	186	1.75	5.00	3.5242	.68088	-.103
Overall BI utilisation (all 9 items)	186	1.22	5.00	3.7838	.66373	-.861
Managerial BI (Strategic + Tactical)	186	1.00	5.00	3.7643	.69264	-.809
Valid N (listwise)	186					

Descriptive Statistics

	Skewness	Kurtosis	
	Std. Error	Statistic	Std. Error
Strategic BI utilisation composite	.178	1.015	.355
Tactical BI utilisation composite	.178	1.943	.355
Operational BI utilisation composite	.178	.507	.355
Decision-making quality composite	.178	.371	.355
BI challenges composite	.178	-.099	.355
Overall BI utilisation (all 9 items)	.178	1.748	.355
Managerial BI (Strategic + Tactical)	.178	1.950	.355
Valid N (listwise)			

```
 * Item-level descriptives.
DESCRIPTIVES VARIABLES=Strategic_BI_Availability TO BI_Data_Integration_Issue
  /STATISTICS=MEAN STDDEV MIN MAX SKEWNESS KURTOSIS.
```

Descriptives

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=Strategic_BI_Availability TO BI_Data_Integration_Issue /STATISTICS=MEAN STDDEV MIN MAX SKEWNESS KURTOSIS.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation	Skewness
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic
Strategic_BI_Availability	186	1.00	5.00	3.6452	.95459	-.705
Strategic_BI_Alternatives	186	1.00	5.00	3.7419	.91714	-.739
Strategic_BI_Alignment	186	1.00	5.00	3.8118	.87120	-.566
Tactical_BI_Availability	186	1.00	5.00	3.7097	.90142	-.556
Tactical_BI_Resource_Alloc	186	1.00	5.00	3.7688	.82874	-.639
Tactical_BI_Data_Decisions	186	1.00	5.00	3.9086	.83000	-.859
Operational_BI_Availability	186	1.00	5.00	3.7903	.95528	-.734
Operational_BI_Agility	186	1.00	5.00	3.8495	.85039	-.559
Operational_BI_Quality	186	1.00	5.00	3.8280	.86523	-.772
BI_Improves_Decisions	186	1.00	5.00	3.8495	.86301	-.571
BI_Speed_Accuracy	186	1.00	5.00	3.7742	.96567	-.589
BI_Data_Inform	186	1.00	5.00	3.9355	.87988	-.836
BI_Alternative_Solutions	186	1.00	5.00	3.7097	.89540	-.261
BI_Cost_Barrier	186	1.00	5.00	3.5215	1.04083	-.479
BI_Training_Limit	186	1.00	5.00	3.7634	.96308	-.463
BI_Flexibility_Issue	186	1.00	5.00	3.3280	.98364	-.182
BI_Data_Integration_Issue	186	1.00	5.00	3.4839	.90778	-.280
Valid N (listwise)	186					

Descriptive Statistics

	Skewness	Kurtosis	
	Std. Error	Statistic	Std. Error
Strategic_BI_Availability	.178	.410	.355
Strategic_BI_Alternatives	.178	.528	.355
Strategic_BI_Alignment	.178	.492	.355
Tactical_BI_Availability	.178	.013	.355
Tactical_BI_Resource_Alloc	.178	.906	.355
Tactical_BI_Data_Decisions	.178	1.395	.355
Operational_BI_Availability	.178	.316	.355
Operational_BI_Agility	.178	.387	.355
Operational_BI_Quality	.178	.858	.355
BI_Improves_Decisions	.178	.079	.355
BI_Speed_Accuracy	.178	-.088	.355
BI_Data_Inform	.178	.691	.355
BI_Alternative_Solutions	.178	-.444	.355
BI_Cost_Barrier	.178	-.211	.355
BI_Training_Limit	.178	-.375	.355
BI_Flexibility_Issue	.178	-.241	.355
BI_Data_Integration_Issue	.178	-.008	.355
Valid N (listwise)			

```
 * -------------------- 4. RELIABILITY (CRONBACH ALPHA) --------------------

RELIABILITY
  /VARIABLES=Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_
Alignment
  /SCALE('Strategic BI') ALL
  /MODEL=ALPHA
  /STATISTICS=DESCRIPTIVE SCALE
  /SUMMARY=TOTAL.
```

Reliability

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY /VARIABLES=Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_Alignment /SCALE('Strategic BI') ALL /MODEL=ALPHA /STATISTICS=DESCRIPTIVE SCALE /SUMMARY=TOTAL.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Scale: Strategic BI

Case Processing Summary

		N	%
Cases	Valid	186	100.0
	Excluded[a]	0	.0
	Total	186	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.800	3

Item Statistics

	Mean	Std. Deviation	N
Strategic_BI_Availability	3.6452	.95459	186
Strategic_BI_Alternatives	3.7419	.91714	186
Strategic_BI_Alignment	3.8118	.87120	186

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Strategic_BI_Availability	7.5538	2.573	.620	.756
Strategic_BI_Alternatives	7.4570	2.498	.705	.663
Strategic_BI_Alignment	7.3871	2.822	.615	.758

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
11.1989	5.382	2.31988	3

```
RELIABILITY
  /VARIABLES=Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_D
ata_Decisions
  /SCALE('Tactical BI') ALL
```

```
/MODEL=ALPHA
/STATISTICS=DESCRIPTIVE SCALE
/SUMMARY=TOTAL.
```

Reliability

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY /VARIABLES=Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_Data_Decisions /SCALE('Tactical BI') ALL /MODEL=ALPHA /STATISTICS=DESCRIPTIVE SCALE /SUMMARY=TOTAL.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Scale: Tactical BI

Case Processing Summary

		N	%
Cases	Valid	186	100.0
	Excluded[a]	0	.0
	Total	186	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.768	3

Item Statistics

	Mean	Std. Deviation	N
Tactical_BI_Availability	3.7097	.90142	186
Tactical_BI_Resource_Alloc	3.7688	.82874	186
Tactical_BI_Data_Decisions	3.9086	.83000	186

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Tactical_BI_Availability	7.6774	2.166	.569	.730
Tactical_BI_Resource_Alloc	7.6183	2.205	.648	.638
Tactical_BI_Data_Decisions	7.4785	2.305	.593	.699

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
11.3871	4.487	2.11830	3

```
RELIABILITY
```

```
  /VARIABLES=Operational_BI_Availability Operational_BI_Agility Operational_BI_Quality
  /SCALE('Operational BI') ALL
  /MODEL=ALPHA
  /STATISTICS=DESCRIPTIVE SCALE
  /SUMMARY=TOTAL.
```

Reliability

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY /VARIABLES=Operational_BI_Availability Operational_BI_Agility Operational_BI_Quality /SCALE('Operational BI') ALL /MODEL=ALPHA /STATISTICS=DESCRIPTIVE SCALE /SUMMARY=TOTAL.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Scale: Operational BI

Case Processing Summary

		N	%
Cases	Valid	186	100.0
	Excluded[a]	0	.0
	Total	186	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.781	3

Item Statistics

	Mean	Std. Deviation	N
Operational_BI_Availability	3.7903	.95528	186
Operational_BI_Agility	3.8495	.85039	186
Operational_BI_Quality	3.8280	.86523	186

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Operational_BI_Availability	7.6774	2.317	.600	.730
Operational_BI_Agility	7.6183	2.518	.642	.681
Operational_BI_Quality	7.6398	2.524	.619	.704

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
11.4677	4.975	2.23039	3

```
RELIABILITY
```

```
  /VARIABLES=BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alterna
tive_Solutions
  /SCALE('DMQ') ALL
  /MODEL=ALPHA
  /STATISTICS=DESCRIPTIVE SCALE
  /SUMMARY=TOTAL.
```

Reliability

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY /VARIABLES=BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions /SCALE('DMQ') ALL /MODEL=ALPHA /STATISTICS=DESCRIPTIVE SCALE /SUMMARY=TOTAL.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Scale: DMQ

Case Processing Summary

		N	%
Cases	Valid	186	100.0
	Excluded[a]	0	.0
	Total	186	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.845	4

Item Statistics

	Mean	Std. Deviation	N
BI_Improves_Decisions	3.8495	.86301	186
BI_Speed_Accuracy	3.7742	.96567	186
BI_Data_Inform	3.9355	.87988	186
BI_Alternative_Solutions	3.7097	.89540	186

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
BI_Improves_Decisions	11.4194	5.434	.674	.808
BI_Speed_Accuracy	11.4946	4.824	.738	.778
BI_Data_Inform	11.3333	5.337	.683	.803
BI_Alternative_Solutions	11.5591	5.437	.635	.824

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
15.2688	8.890	2.98153	4

```
RELIABILITY
  /VARIABLES=BI_Cost_Barrier BI_Training_Limit BI_Flexibility_Issue BI_Data_In
tegration_Issue
  /SCALE('Challenges') ALL
  /MODEL=ALPHA
  /STATISTICS=DESCRIPTIVE SCALE
  /SUMMARY=TOTAL.
```

Reliability

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.
Syntax		RELIABILITY /VARIABLES=BI_Cost_Barrier BI_Training_Limit BI_Flexibility_Issue BI_Data_Integration_Issue /SCALE('Challenges') ALL /MODEL=ALPHA /STATISTICS=DESCRIPTIVE SCALE /SUMMARY=TOTAL.

Notes

Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Scale: Challenges

Case Processing Summary

		N	%
Cases	Valid	186	100.0
	Excluded[a]	0	.0
	Total	186	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.650	4

Item Statistics

	Mean	Std. Deviation	N
BI_Cost_Barrier	3.5215	1.04083	186
BI_Training_Limit	3.7634	.96308	186
BI_Flexibility_Issue	3.3280	.98364	186
BI_Data_Integration_Issue	3.4839	.90778	186

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
BI_Cost_Barrier	10.5753	4.711	.359	.634
BI_Training_Limit	10.3333	4.645	.444	.572
BI_Flexibility_Issue	10.7688	4.557	.451	.567
BI_Data_Integration_Issue	10.6129	4.725	.473	.554

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
14.0968	7.418	2.72353	4

```
RELIABILITY
  /VARIABLES=Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_
Alignment
             Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_D
ata_Decisions
             Operational_BI_Availability Operational_BI_Agility Operational_BI
_Quality
  /SCALE('Overall BI') ALL
  /MODEL=ALPHA
  /STATISTICS=DESCRIPTIVE SCALE
  /SUMMARY=TOTAL.
```

Reliability

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.

Notes

Syntax		RELIABILITY /VARIABLES=Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_Alignment Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_Data_Decisions Operational_BI_Availability Operational_BI_Agility Operational_BI_Quality /SCALE('Overall BI') ALL /MODEL=ALPHA /STATISTICS=DESCRIPTIVE SCALE /SUMMARY=TOTAL.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.02

Scale: Overall BI

Case Processing Summary

		N	%
Cases	Valid	186	100.0
	Excluded[a]	0	.0
	Total	186	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.902	9

Item Statistics

	Mean	Std. Deviation	N
Strategic_BI_Availability	3.6452	.95459	186
Strategic_BI_Alternatives	3.7419	.91714	186
Strategic_BI_Alignment	3.8118	.87120	186
Tactical_BI_Availability	3.7097	.90142	186
Tactical_BI_Resource_Alloc	3.7688	.82874	186
Tactical_BI_Data_Decisions	3.9086	.83000	186
Operational_BI_Availability	3.7903	.95528	186
Operational_BI_Agility	3.8495	.85039	186
Operational_BI_Quality	3.8280	.86523	186

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Strategic_BI_Availability	30.4086	28.254	.642	.893
Strategic_BI_Alternatives	30.3118	27.686	.741	.885
Strategic_BI_Alignment	30.2419	28.660	.672	.891
Tactical_BI_Availability	30.3441	28.259	.690	.889
Tactical_BI_Resource_Alloc	30.2849	28.756	.702	.889
Tactical_BI_Data_Decisions	30.1452	29.206	.645	.893
Operational_BI_Availability	30.2634	28.206	.647	.893
Operational_BI_Agility	30.2043	29.115	.637	.893
Operational_BI_Quality	30.2258	28.727	.669	.891

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
34.0538	35.684	5.97357	9

RELIABILITY

```
  /VARIABLES=Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_
Alignment
             Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_D
ata_Decisions
  /SCALE('Managerial BI') ALL
  /MODEL=ALPHA
  /STATISTICS=DESCRIPTIVE SCALE
  /SUMMARY=TOTAL.
```

Reliability

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
	Matrix Input	
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on all cases with valid data for all variables in the procedure.

Notes

Syntax		RELIABILITY /VARIABLES=Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_Alignment Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_Data_Decisions /SCALE('Managerial BI') ALL /MODEL=ALPHA /STATISTICS=DESCRIPTIVE SCALE /SUMMARY=TOTAL.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Scale: Managerial BI

Case Processing Summary

		N	%
Cases	Valid	186	100.0
	Excluded[a]	0	.0
	Total	186	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.873	6

Item Statistics

	Mean	Std. Deviation	N
Strategic_BI_Availability	3.6452	.95459	186
Strategic_BI_Alternatives	3.7419	.91714	186
Strategic_BI_Alignment	3.8118	.87120	186
Tactical_BI_Availability	3.7097	.90142	186
Tactical_BI_Resource_Alloc	3.7688	.82874	186
Tactical_BI_Data_Decisions	3.9086	.83000	186

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Strategic_BI_Availability	18.9409	12.132	.636	.860
Strategic_BI_Alternatives	18.8441	11.635	.766	.835
Strategic_BI_Alignment	18.7742	12.392	.672	.852
Tactical_BI_Availability	18.8763	12.217	.673	.852
Tactical_BI_Resource_Alloc	18.8172	12.561	.685	.851
Tactical_BI_Data_Decisions	18.6774	12.847	.628	.860

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
22.5860	17.271	4.15583	6

```
* -------------------- 5. EFA (PAF, PROMAX) ON 13 BI + DMQ ITEMS --------------------

* 5a. KMO and Bartlett.
* 5b. Principal Axis Factoring with Promax rotation.
* 5c. Primary regression uses standardized composite mean scores (Section 9), not factor scores.
```

```
FACTOR
  /VARIABLES Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_
Alignment
             Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_D
ata_Decisions
             Operational_BI_Availability Operational_BI_Agility Operational_BI
_Quality
             BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alterna
tive_Solutions
  /MISSING LISTWISE
  /ANALYSIS Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_A
lignment
             Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_D
ata_Decisions
             Operational_BI_Availability Operational_BI_Agility Operational_BI
_Quality
             BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alterna
tive_Solutions
  /PRINT INITIAL KMO REPR EXTRACTION ROTATION
  /FORMAT SORT BLANK(.30)
  /CRITERIA FACTORS(4) ITERATE(25)
  /EXTRACTION PAF
  /CRITERIA ITERATE(25)
  /ROTATION PROMAX(4)
  /METHOD=CORRELATION.
```

Factor Analysis

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.

Notes

Syntax	FACTOR
	  /VARIABLES Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_Alignment
	
	Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_Data_Decisions
	
	Operational_BI_Availability Operational_BI_Agility Operational_BI_Quality
	
	BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions
	  /MISSING LISTWISE
	  /ANALYSIS Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_Alignment
	
	Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_Data_Decisions
	
	Operational_BI_Availability Operational_BI_Agility Operational_BI_Quality
	
	BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions
	  /PRINT INITIAL KMO REPR EXTRACTION ROTATION
	  /FORMAT SORT BLANK (.30)
	  /CRITERIA FACTORS(4) ITERATE(25)
	  /EXTRACTION PAF
	  /CRITERIA ITERATE(25)
	  /ROTATION PROMAX(4)
	
	/METHOD=CORRELATION.

Notes

Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.02
	Maximum Memory Required	21700 (21.191K) bytes

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.936
Bartlett's Test of Sphericity	Approx. Chi-Square	1330.530
	df	78
	Sig.	.000

Communalities

	Initial	Extraction
Strategic_BI_Availability	.483	.478
Strategic_BI_Alternatives	.606	.718
Strategic_BI_Alignment	.540	.544
Tactical_BI_Availability	.563	.816
Tactical_BI_Resource_Alloc	.529	.548
Tactical_BI_Data_Decisions	.479	.491
Operational_BI_Availability	.485	.498
Operational_BI_Agility	.520	.648
Operational_BI_Quality	.500	.540
BI_Improves_Decisions	.600	.630
BI_Speed_Accuracy	.604	.653
BI_Data_Inform	.546	.588
BI_Alternative_Solutions	.521	.767

Extraction Method: Principal Axis Factoring.

Total Variance Explained

Factor	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	7.074	54.417	54.417	6.687	51.441	51.441
2	.848	6.521	60.938	.505	3.881	55.322
3	.749	5.764	66.702	.451	3.467	58.790
4	.654	5.030	71.732	.276	2.126	60.916
5	.625	4.805	76.536			
6	.562	4.326	80.862			
7	.521	4.009	84.871			
8	.417	3.207	88.078			
9	.369	2.841	90.919			
10	.344	2.649	93.568			
11	.308	2.370	95.937			
12	.272	2.093	98.030			
13	.256	1.970	100.000			

Total Variance Explained

Factor	Rotation Sums of Squared Loadings[a]
	Total
1	6.048
2	5.710
3	4.498
4	2.558
5	
6	
7	
8	
9	
10	
11	
12	
13	

Extraction Method: Principal Axis Factoring.

a. When factors are correlated, sums of squared loadings cannot be added to obtain a total variance.

Factor Matrix[a]

	Factor			
	1	2	3	4
BI_Improves_Decisions	.778			
Strategic_BI_Alternatives	.774			
BI_Speed_Accuracy	.744	.309		
Tactical_BI_Availability	.743	-.349		
Tactical_BI_Resource_Alloc	.732			
BI_Data_Inform	.721			
Operational_BI_Agility	.714			
Strategic_BI_Alignment	.708			
Operational_BI_Quality	.698			
BI_Alternative_Solutions	.686		-.522	
Operational_BI_Availability	.676			
Tactical_BI_Data_Decisions	.673			
Strategic_BI_Availability	.664			

Extraction Method: Principal Axis Factoring.

a. Attempted to extract 4 factors. More than 25 iterations required. (Convergence=.001). Extraction was terminated.

Reproduced Correlations

		Strategic_BI_Availability	Strategic_BI_Alternatives	Strategic_BI_Alignment
Reproduced Correlation	Strategic_BI_Availability	.478[a]	.569	.497
	Strategic_BI_Alternatives	.569	.718[a]	.618
	Strategic_BI_Alignment	.497	.618	.544[a]
	Tactical_BI_Availability	.525	.603	.529
	Tactical_BI_Resource_Alloc	.503	.606	.541
	Tactical_BI_Data_Decisions	.475	.569	.504
	Operational_BI_Availability	.454	.492	.453
	Operational_BI_Agility	.444	.452	.441
	Operational_BI_Quality	.470	.507	.466
	BI_Improves_Decisions	.507	.588	.546
	BI_Speed_Accuracy	.438	.495	.488
	BI_Data_Inform	.441	.527	.503
	BI_Alternative_Solutions	.382	.464	.461
Residual[b]	Strategic_BI_Availability		.042	.001
	Strategic_BI_Alternatives	.042		-.009
	Strategic_BI_Alignment	.001	-.009	
	Tactical_BI_Availability	.015	-.007	-.007
	Tactical_BI_Resource_Alloc	-.041	-.009	.004
	Tactical_BI_Data_Decisions	-.053	-.004	-.012
	Operational_BI_Availability	-.032	.026	-.046
	Operational_BI_Agility	-.010	-.003	.009
	Operational_BI_Quality	.025	-.032	.065
	BI_Improves_Decisions	.025	-.010	-.001
	BI_Speed_Accuracy	.049	.025	-.070
	BI_Data_Inform	-.011	-.019	.059
	BI_Alternative_Solutions	-.022	.004	.009

Reproduced Correlations

		Tactical_BI_Availability	Tactical_BI_Resource_Alloc	Tactical_BI_Data_Decisions
Reproduced Correlation	Strategic_BI_Availability	.525	.503	.475
	Strategic_BI_Alternatives	.603	.606	.569
	Strategic_BI_Alignment	.529	.541	.504
	Tactical_BI_Availability	.816[a]	.546	.463
	Tactical_BI_Resource_Alloc	.546	.548[a]	.510
	Tactical_BI_Data_Decisions	.463	.510	.491[a]
	Operational_BI_Availability	.506	.484	.457
	Operational_BI_Agility	.450	.491	.469
	Operational_BI_Quality	.523	.499	.473
	BI_Improves_Decisions	.499	.567	.534
	BI_Speed_Accuracy	.446	.520	.473
	BI_Data_Inform	.434	.519	.475
	BI_Alternative_Solutions	.582	.478	.373
Residual[b]	Strategic_BI_Availability	.015	-.041	-.053
	Strategic_BI_Alternatives	-.007	-.009	-.004
	Strategic_BI_Alignment	-.007	.004	-.012
	Tactical_BI_Availability		-.007	.007
	Tactical_BI_Resource_Alloc	-.007		.065
	Tactical_BI_Data_Decisions	.007	.065	
	Operational_BI_Availability	.013	.041	.009
	Operational_BI_Agility	-.007	-.011	.017
	Operational_BI_Quality	-.012	-.005	-.036
	BI_Improves_Decisions	.007	-.034	.043
	BI_Speed_Accuracy	-.001	-.005	-.027
	BI_Data_Inform	-.001	-.006	-.017
	BI_Alternative_Solutions	.003	.014	-.001

Reproduced Correlations

		Operational_BI_Availability	Operational_BI_Agility	Operational_BI_Quality
Reproduced Correlation	Strategic_BI_Availability	.454	.444	.470
	Strategic_BI_Alternatives	.492	.452	.507
	Strategic_BI_Alignment	.453	.441	.466
	Tactical_BI_Availability	.506	.450	.523
	Tactical_BI_Resource_Alloc	.484	.491	.499
	Tactical_BI_Data_Decisions	.457	.469	.473
	Operational_BI_Availability	.498[a]	.543	.518
	Operational_BI_Agility	.543	.648[a]	.566
	Operational_BI_Quality	.518	.566	.540[a]
	BI_Improves_Decisions	.531	.589	.548
	BI_Speed_Accuracy	.501	.598	.516
	BI_Data_Inform	.464	.531	.475
	BI_Alternative_Solutions	.401	.434	.405
Residual[b]	Strategic_BI_Availability	-.032	-.010	.025
	Strategic_BI_Alternatives	.026	-.003	-.032
	Strategic_BI_Alignment	-.046	.009	.065
	Tactical_BI_Availability	.013	-.007	-.012
	Tactical_BI_Resource_Alloc	.041	-.011	-.005
	Tactical_BI_Data_Decisions	.009	.017	-.036
	Operational_BI_Availability		.004	.000
	Operational_BI_Agility	.004		.009
	Operational_BI_Quality	.000	.009	
	BI_Improves_Decisions	-.051	.006	.010
	BI_Speed_Accuracy	.009	-.014	-.013
	BI_Data_Inform	.041	-.009	-.014
	BI_Alternative_Solutions	-.018	.012	.005

Reproduced Correlations

		BI_Improves_Decisions	BI_Speed_Accuracy	BI_Data_Inform
Reproduced Correlation	Strategic_BI_Availability	.507	.438	.441
	Strategic_BI_Alternatives	.588	.495	.527
	Strategic_BI_Alignment	.546	.488	.503
	Tactical_BI_Availability	.499	.446	.434
	Tactical_BI_Resource_Alloc	.567	.520	.519
	Tactical_BI_Data_Decisions	.534	.473	.475
	Operational_BI_Availability	.531	.501	.464
	Operational_BI_Agility	.589	.598	.531
	Operational_BI_Quality	.548	.516	.475
	BI_Improves_Decisions	.630[a]	.612	.589
	BI_Speed_Accuracy	.612	.653[a]	.609
	BI_Data_Inform	.589	.609	.588[a]
	BI_Alternative_Solutions	.506	.582	.556
Residual[b]	Strategic_BI_Availability	.025	.049	-.011
	Strategic_BI_Alternatives	-.010	.025	-.019
	Strategic_BI_Alignment	-.001	-.070	.059
	Tactical_BI_Availability	.007	-.001	-.001
	Tactical_BI_Resource_Alloc	-.034	-.005	-.006
	Tactical_BI_Data_Decisions	.043	-.027	-.017
	Operational_BI_Availability	-.051	.009	.041
	Operational_BI_Agility	.006	-.014	-.009
	Operational_BI_Quality	.010	-.013	-.014
	BI_Improves_Decisions		.028	-.025
	BI_Speed_Accuracy	.028		.017
	BI_Data_Inform	-.025	.017	
	BI_Alternative_Solutions	.004	-.001	-.011

Reproduced Correlations

		BI_Alternative_Solutions
Reproduced Correlation	Strategic_BI_Availability	.382
	Strategic_BI_Alternatives	.464
	Strategic_BI_Alignment	.461
	Tactical_BI_Availability	.582
	Tactical_BI_Resource_Alloc	.478
	Tactical_BI_Data_Decisions	.373
	Operational_BI_Availability	.401
	Operational_BI_Agility	.434
	Operational_BI_Quality	.405
	BI_Improves_Decisions	.506
	BI_Speed_Accuracy	.582
	BI_Data_Inform	.556
	BI_Alternative_Solutions	.767[a]
Residual[b]	Strategic_BI_Availability	-.022
	Strategic_BI_Alternatives	.004
	Strategic_BI_Alignment	.009
	Tactical_BI_Availability	.003
	Tactical_BI_Resource_Alloc	.014
	Tactical_BI_Data_Decisions	-.001
	Operational_BI_Availability	-.018
	Operational_BI_Agility	.012
	Operational_BI_Quality	.005
	BI_Improves_Decisions	.004
	BI_Speed_Accuracy	-.001
	BI_Data_Inform	-.011
	BI_Alternative_Solutions	

Extraction Method: Principal Axis Factoring.

a. Reproduced communalities

b. Residuals are computed between observed and reproduced correlations. There are 6 (7.0%) nonredundant residuals with absolute values greater than 0.05.

Pattern Matrix[a]

	Factor			
	1	2	3	4
Strategic_BI_Alternatives	.934			
Strategic_BI_Alignment	.682			
Tactical_BI_Data_Decisions	.594			
Strategic_BI_Availability	.541			
Tactical_BI_Resource_Alloc	.531			
Operational_BI_Agility		.891		
Operational_BI_Quality		.618		
Operational_BI_Availability		.570		
BI_Speed_Accuracy		.491	.392	
BI_Improves_Decisions	.371	.405		
BI_Alternative_Solutions			.893	
BI_Data_Inform			.374	
Tactical_BI_Availability				.673

Extraction Method: Principal Axis Factoring.
Rotation Method: Promax with Kaiser Normalization.

a. Rotation converged in 9 iterations.

Structure Matrix

	Factor			
	1	2	3	4
Strategic_BI_Alternatives	.840	.612	.541	.499
Strategic_BI_Alignment	.735	.587	.544	.416
Tactical_BI_Resource_Allo c	.730	.641	.562	.433
Tactical_BI_Data_Decision s	.690	.613	.459	.347
Strategic_BI_Availability	.677	.581	.440	.450
Operational_BI_Agility	.597	.802	.541	.310
BI_Improves_Decisions	.736	.754	.628	.333
BI_Speed_Accuracy	.644	.753	.721	
Operational_BI_Quality	.632	.712	.475	.435
Operational_BI_Availability	.613	.684	.471	.419
BI_Alternative_Solutions	.571	.554	.855	.461
BI_Data_Inform	.667	.681	.692	
Tactical_BI_Availability	.691	.583	.570	.848

Extraction Method: Principal Axis Factoring.
Rotation Method: Promax with Kaiser Normalization.

Factor Correlation Matrix

Factor	1	2	3	4
1	1.000	.795	.683	.537
2	.795	1.000	.692	.401
3	.683	.692	1.000	.366
4	.537	.401	.366	1.000

Extraction Method: Principal Axis Factoring.
Rotation Method: Promax with Kaiser Normalization.

```
 * Note: Factor scores are not saved. Primary regression uses standardized
* composite mean scores (see Section 9).

* -------------------- 6. HARMAN SINGLE-FACTOR TEST (CMV) --------------------
```

```
FACTOR
  /VARIABLES Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_
Alignment
             Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_D
ata_Decisions
             Operational_BI_Availability Operational_BI_Agility Operational_BI
_Quality
             BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alterna
tive_Solutions
  /MISSING LISTWISE
  /ANALYSIS Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_A
lignment
             Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_D
ata_Decisions
             Operational_BI_Availability Operational_BI_Agility Operational_BI
_Quality
             BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alterna
tive_Solutions
  /PRINT INITIAL EXTRACTION
  /CRITERIA FACTORS(1) ITERATE(25)
  /EXTRACTION PAF
  /ROTATION NOROTATE
  /METHOD=CORRELATION.
```

Factor Analysis

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	MISSING=EXCLUDE: User-defined missing values are treated as missing.
	Cases Used	LISTWISE: Statistics are based on cases with no missing values for any variable used.

Notes

Syntax	FACTOR
	  /VARIABLES Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_Alignment
	
	Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_Data_Decisions
	
	Operational_BI_Availability Operational_BI_Agility Operational_BI_Quality
	
	BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions
	  /MISSING LISTWISE
	  /ANALYSIS Strategic_BI_Availability Strategic_BI_Alternatives Strategic_BI_Alignment
	
	Tactical_BI_Availability Tactical_BI_Resource_Alloc Tactical_BI_Data_Decisions
	
	Operational_BI_Availability Operational_BI_Agility Operational_BI_Quality
	
	BI_Improves_Decisions BI_Speed_Accuracy BI_Data_Inform BI_Alternative_Solutions
	  /PRINT INITIAL EXTRACTION
	  /CRITERIA FACTORS(1) ITERATE(25)
	  /EXTRACTION PAF
	  /ROTATION NOROTATE
	
	/METHOD=CORRELATION.

Notes

Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.02
	Maximum Memory Required	21700 (21.191K) bytes

Communalities

	Initial	Extraction
Strategic_BI_Availability	.483	.443
Strategic_BI_Alternatives	.606	.586
Strategic_BI_Alignment	.540	.503
Tactical_BI_Availability	.563	.510
Tactical_BI_Resource_Alloc	.529	.543
Tactical_BI_Data_Decisions	.479	.456
Operational_BI_Availability	.485	.459
Operational_BI_Agility	.520	.495
Operational_BI_Quality	.500	.487
BI_Improves_Decisions	.600	.612
BI_Speed_Accuracy	.604	.544
BI_Data_Inform	.546	.516
BI_Alternative_Solutions	.521	.431

Extraction Method: Principal Axis Factoring.

Total Variance Explained

Factor	Initial Eigenvalues Total	Initial Eigenvalues % of Variance	Initial Eigenvalues Cumulative %	Extraction Sums of Squared Loadings Total	Extraction Sums of Squared Loadings % of Variance	Extraction Sums of Squared Loadings Cumulative %
1	7.074	54.417	54.417	6.586	50.659	50.659
2	.848	6.521	60.938			
3	.749	5.764	66.702			
4	.654	5.030	71.732			
5	.625	4.805	76.536			
6	.562	4.326	80.862			
7	.521	4.009	84.871			
8	.417	3.207	88.078			
9	.369	2.841	90.919			
10	.344	2.649	93.568			
11	.308	2.370	95.937			
12	.272	2.093	98.030			
13	.256	1.970	100.000			

Extraction Method: Principal Axis Factoring.

Factor Matrix[a]

	Factor 1
Strategic_BI_Availability	.666
Strategic_BI_Alternatives	.765
Strategic_BI_Alignment	.709
Tactical_BI_Availability	.714
Tactical_BI_Resource_Alloc	.737
Tactical_BI_Data_Decisions	.675
Operational_BI_Availability	.678
Operational_BI_Agility	.703
Operational_BI_Quality	.698
BI_Improves_Decisions	.782
BI_Speed_Accuracy	.738
BI_Data_Inform	.719
BI_Alternative_Solutions	.656

Extraction Method: Principal Axis Factoring.

a. 1 factors extracted. 4 iterations required.

```
* -------------------- 7. CORRELATIONS --------------------

* 7a. Pearson correlations among composites.
CORRELATIONS
  /VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean DMQ_Mean
             Overall_BI_Mean Managerial_BI_Mean Challenges_Mean
  /PRINT=TWOTAIL NOSIG FULL
  /MISSING=PAIRWISE.
```

Correlations

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Syntax		CORRELATIONS /VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean DMQ_Mean Overall_BI_Mean Managerial_BI_Mean Challenges_Mean /PRINT=TWOTAIL NOSIG FULL /MISSING=PAIRWISE.

Notes

Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.00

Warnings

There is an invalid keyword on the PRINT subcommand. Text found: FULL
Execution of this command stops.

```
* 7b. Spearman correlations (robustness for ordinal data).
NONPAR CORR
  /VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean DMQ_Mean
             Overall_BI_Mean Managerial_BI_Mean Challenges_Mean
  /PRINT=SPEARMAN TWOTAIL NOSIG FULL
  /MISSING=PAIRWISE.
```

Nonparametric Correlations

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each pair of variables are based on all the cases with valid data for that pair.
Syntax		NONPAR CORR /VARIABLES=Strategic_Mean Tactical_Mean Operational_Mean DMQ_Mean Overall_BI_Mean Managerial_BI_Mean Challenges_Mean /PRINT=SPEARMAN TWOTAIL NOSIG FULL /MISSING=PAIRWISE.
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.02
	Number of Cases Allowed	8017796 cases[a]

a. Based on availability of workspace memory

Warnings

There is an invalid keyword on the PRINT subcommand. Text found: "FULL".
Execution of this command stops.

```
* -------------------- 8. CONTROL VARIABLE DUMMY CODING --------------------

* Company_Size: reference = 1 (<50 employees).
RECODE Company_Size (2=1)(ELSE=0) INTO CS_50_99.
RECODE Company_Size (3=1)(ELSE=0) INTO CS_100_249.

* Job_Position: reference = 1 (CEO).
RECODE Job_Position (2=1)(ELSE=0) INTO JP_SeniorMgr.
RECODE Job_Position (3=1)(ELSE=0) INTO JP_MiddleMgr.
RECODE Job_Position (4=1)(ELSE=0) INTO JP_OpsMgr.
RECODE Job_Position (5=1)(ELSE=0) INTO JP_Other.

* Industry: reference = 1 (Manufacturing).
RECODE Industry (2=1)(ELSE=0) INTO IND_Retail.
RECODE Industry (3=1)(ELSE=0) INTO IND_Service.
RECODE Industry (4=1)(ELSE=0) INTO IND_Technology.
RECODE Industry (5=1)(ELSE=0) INTO IND_Other.

EXECUTE.

VARIABLE LABELS
  CS_50_99 'Company size 50-99 (ref: <50)'
  CS_100_249 'Company size 100-249 (ref: <50)'
  JP_SeniorMgr 'Job: Senior Manager (ref: CEO)'
  JP_MiddleMgr 'Job: Middle Manager (ref: CEO)'
  JP_OpsMgr 'Job: Operational Manager (ref: CEO)'
  JP_Other 'Job: Other (ref: CEO)'
  IND_Retail 'Industry: Retail (ref: Manufacturing)'
  IND_Service 'Industry: Service (ref: Manufacturing)'
  IND_Technology 'Industry: Technology (ref: Manufacturing)'
  IND_Other 'Industry: Other (ref: Manufacturing)'.

* -------------------- 9. MAIN REGRESSION (Composite means -> DMQ) -----------
---------

* Model M1: Controls only.
REGRESSION
  /MISSING LISTWISE
  /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI(95)
  /CRITERIA=PIN(.05) POUT(.10)
  /NOORIGIN
  /DEPENDENT DMQ_Mean
```

```
  /METHOD=ENTER CS_50_99 CS_100_249 JP_SeniorMgr JP_MiddleMgr JP_OpsMgr JP_Oth
er
                IND_Retail IND_Service IND_Technology IND_Other
  /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID)
  /CASEWISE PLOT(ZRESID) OUTLIERS(3)
  /SAVE COOK LEVER ZRESID.
```

Regression

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.

Notes

Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI (95) /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT DMQ_Mean /METHOD=ENTER CS_50_99 CS_100_249 JP_SeniorMgr JP_MiddleMgr JP_OpsMgr JP_Other IND_Retail IND_Service IND_Technology IND_Other /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID) /CASEWISE PLOT (ZRESID) OUTLIERS(3) /SAVE COOK LEVER ...
Resources	Processor Time	00:00:00.17
	Elapsed Time	00:00:00.22
	Memory Required	5628 bytes
	Additional Memory Required for Residual Plots	584 bytes
Variables Created or Modified	ZRE_1	Standardized Residual
	COO_1	Cook's Distance
	LEV_1	Centered Leverage Value

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Industry: Other (ref: Manufacturing), Job: Middle Manager (ref: CEO), Company size 100-249 (ref: <50), Job: Senior Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Industry: Service (ref: Manufacturing), Company size 50-99 (ref: <50), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)[b]	.	Enter

a. Dependent Variable: Decision-making quality composite

b. All requested variables entered.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics		
					R Square Change	F Change	df1
1	.350[a]	.123	.073	.71780	.123	2.449	10

Model Summary[b]

Model	Change Statistics: df2	Sig. F Change	Durbin-Watson
1	175	.009	1.835

a. Predictors: (Constant), Industry: Other (ref: Manufacturing), Job: Middle Manager (ref: CEO), Company size 100-249 (ref: <50), Job: Senior Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Industry: Service (ref: Manufacturing), Company size 50-99 (ref: <50), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)

b.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	12.619	10	1.262	2.449	.009[b]
	Residual	90.166	175	.515		
	Total	102.785	185			

a. Dependent Variable: Decision-making quality composite

b. Predictors: (Constant), Industry: Other (ref: Manufacturing), Job: Middle Manager (ref: CEO), Company size 100-249 (ref: <50), Job: Senior Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Industry: Service (ref: Manufacturing), Company size 50-99 (ref: <50), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	3.832	.261		14.663	.000
	Company size 50-99 (ref: <50)	.321	.132	.210	2.429	.016
	Company size 100-249 (ref: <50)	.145	.143	.087	1.011	.313
	Job: Senior Manager (ref: CEO)	.296	.259	.129	1.142	.255
	Job: Middle Manager (ref: CEO)	.176	.243	.095	.725	.469
	Job: Operational Manager (ref: CEO)	.313	.238	.166	1.314	.190
	Job: Other (ref: CEO)	.130	.233	.087	.559	.577
	Industry: Retail (ref: Manufacturing)	-.588	.231	-.353	-2.548	.012
	Industry: Service (ref: Manufacturing)	-.335	.224	-.203	-1.494	.137
	Industry: Technology (ref: Manufacturing)	-.163	.241	-.082	-.676	.500
	Industry: Other (ref: Manufacturing)	-.402	.241	-.216	-1.672	.096

Coefficients[a]

Model		95.0% Confidence Interval for B		Collinearity Statistics	
		Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	3.316	4.348		
	Company size 50-99 (ref: <50)	.060	.582	.669	1.495
	Company size 100-249 (ref: <50)	-.138	.427	.681	1.467
	Job: Senior Manager (ref: CEO)	-.215	.807	.396	2.525
	Job: Middle Manager (ref: CEO)	-.303	.655	.295	3.390
	Job: Operational Manager (ref: CEO)	-.157	.782	.314	3.188
	Job: Other (ref: CEO)	-.330	.591	.209	4.779
	Industry: Retail (ref: Manufacturing)	-1.043	-.132	.262	3.822
	Industry: Service (ref: Manufacturing)	-.777	.107	.270	3.697
	Industry: Technology (ref: Manufacturing)	-.638	.312	.344	2.907
	Industry: Other (ref: Manufacturing)	-.877	.073	.300	3.330

a. Dependent Variable: Decision-making quality composite

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Company size 50-99 (ref: <50)	Company size 100-249 (ref: <50)
1	1	3.638	1.000	.00	.01	.01
	2	1.515	1.550	.00	.05	.06
	3	1.218	1.728	.00	.00	.00
	4	1.037	1.873	.00	.00	.00
	5	.978	1.929	.00	.00	.00
	6	.954	1.953	.00	.04	.08
	7	.774	2.168	.00	.00	.01
	8	.558	2.553	.00	.09	.23
	9	.246	3.843	.01	.81	.54
	10	.055	8.165	.00	.00	.05
	11	.028	11.479	.99	.00	.01

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions Job: Senior Manager (ref: CEO)	Job: Middle Manager (ref: CEO)	Job: Operational Manager (ref: CEO)	Job: Other (ref: CEO)	Industry: Retail (ref: Manufacturing)
1	1	.00	.00	.00	.00	.00
	2	.00	.00	.04	.02	.00
	3	.01	.06	.01	.01	.03
	4	.11	.02	.00	.00	.04
	5	.16	.03	.01	.00	.03
	6	.01	.01	.00	.00	.01
	7	.01	.07	.07	.00	.03
	8	.01	.01	.12	.05	.00
	9	.02	.03	.01	.00	.01
	10	.34	.40	.41	.60	.53
	11	.33	.38	.33	.31	.32

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions: Industry: Service (ref: Manufacturing)	Industry: Technology (ref: Manufacturing)	Industry: Other (ref: Manufacturing)
1	1	.00	.00	.00
	2	.00	.02	.01
	3	.04	.00	.00
	4	.01	.04	.05
	5	.01	.03	.02
	6	.01	.09	.08
	7	.06	.03	.01
	8	.01	.02	.01
	9	.01	.07	.03
	10	.55	.35	.51
	11	.29	.34	.26

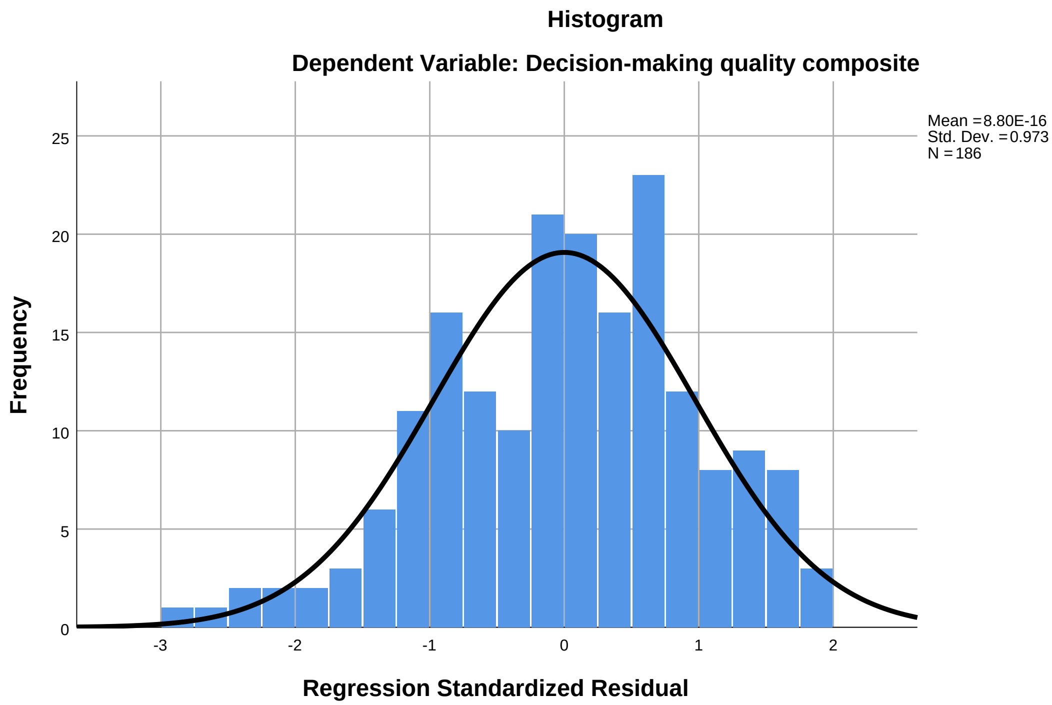
a. Dependent Variable: Decision-making quality composite

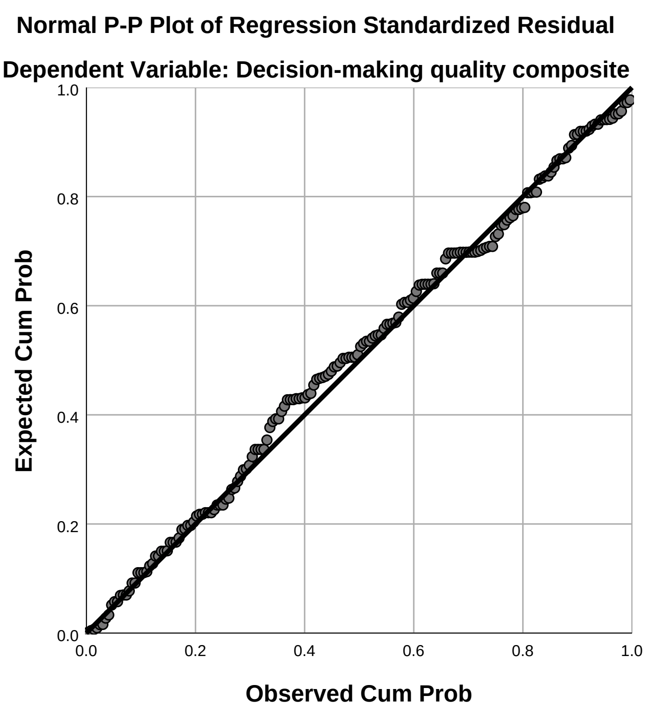
Residuals Statistics[a]

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	3.2442	4.4487	3.8172	.26117	186
Std. Predicted Value	-2.194	2.418	.000	1.000	186
Standard Error of Predicted Value	.134	.275	.171	.033	186
Adjusted Predicted Value	3.2147	4.5121	3.8173	.26397	186
Residual	-2.12724	1.43478	.00000	.69813	186
Std. Residual	-2.964	1.999	.000	.973	186
Stud. Residual	-3.023	2.122	.000	1.001	186
Deleted Residual	-2.21333	1.61651	-.00010	.74044	186
Stud. Deleted Residual	-3.096	2.143	-.001	1.007	186
Mahal. Distance	5.409	26.222	9.946	4.520	186
Cook's Distance	.000	.078	.006	.009	186
Centered Leverage Value	.029	.142	.054	.024	186

a. Dependent Variable: Decision-making quality composite

Charts


Histogram
Dependent Variable: Decision-making quality composite
Mean =8.80E-16
Std. Dev. =0.973
N =186
Frequency
0
5
10
15
20
25
-3
-2
-1
0
1
2
Regression Standardized Residual


Normal P-P Plot of Regression Standardized Residual
Dependent Variable: Decision-making quality composite
Expected Cum Prob
0.0
0.2
0.4
0.6
0.8
1.0
Observed Cum Prob

```
 * Model M2: Three BI levels -> DMQ (PRIMARY MODEL, no controls).
REGRESSION
  /MISSING LISTWISE
  /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI(95) ZPP
  /CRITERIA=PIN(.05) POUT(.10)
  /NOORIGIN
  /DEPENDENT DMQ_Mean
  /METHOD=ENTER Strategic_Mean Tactical_Mean Operational_Mean
  /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID)
  /CASEWISE PLOT(ZRESID) OUTLIERS(3)
  /SCATTERPLOT=(*ZRESID ,*ZPRED).
```

Regression

Notes

Output Created		18-JUN-2026 22:32:47
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.

Notes

Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI (95) ZPP /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT DMQ_Mean /METHOD=ENTER Strategic_Mean Tactical_Mean Operational_Mean /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID) /CASEWISE PLOT (ZRESID) OUTLIERS(3) /SCATTERPLOT=...
Resources	Processor Time	00:00:00.28
	Elapsed Time	00:00:00.54
	Memory Required	2716 bytes
	Additional Memory Required for Residual Plots	896 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Operational BI utilisation composite, Strategic BI utilisation composite, Tactical BI utilisation composite[b]	.	Enter

a. Dependent Variable: Decision-making quality composite

b. All requested variables entered.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics: R Square Change	F Change	df1
1	.794[a]	.630	.624	.45723	.630	103.215	3

Model Summary[b]

Model	Change Statistics: df2	Sig. F Change	Durbin-Watson
1	182	.000	2.264

a. Predictors: (Constant), Operational BI utilisation composite, Strategic BI utilisation composite, Tactical BI utilisation composite

b. Dependent Variable: Decision-making quality composite

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	64.735	3	21.578	103.215	.000[b]
	Residual	38.050	182	.209		
	Total	102.785	185			

a. Dependent Variable: Decision-making quality composite

b. Predictors: (Constant), Operational BI utilisation composite, Strategic BI utilisation composite, Tactical BI utilisation composite

Coefficients[a]

Model		Unstandardized Coefficients: B	Std. Error	Standardized Coefficients: Beta	t	Sig.
1	(Constant)	.437	.196		2.227	.027
	Strategic BI utilisation composite	.254	.069	.264	3.702	.000
	Tactical BI utilisation composite	.278	.080	.263	3.486	.001
	Operational BI utilisation composite	.360	.066	.359	5.451	.000

Coefficients[a]

Model		95.0% Confidence Interval for B Lower Bound	Upper Bound	Correlations Zero-order	Partial	Part
1	(Constant)	.050	.824			
	Strategic BI utilisation composite	.119	.390	.698	.265	.167
	Tactical BI utilisation composite	.121	.435	.714	.250	.157
	Operational BI utilisation composite	.230	.490	.717	.375	.246

Coefficients[a]

Model		Collinearity Statistics Tolerance	VIF
1	(Constant)		
	Strategic BI utilisation composite	.401	2.496
	Tactical BI utilisation composite	.357	2.804
	Operational BI utilisation composite	.468	2.135

a. Dependent Variable: Decision-making quality composite

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Strategic BI utilisation composite	Tactical BI utilisation composite
1	1	3.956	1.000	.00	.00	.00
	2	.023	13.189	.93	.13	.03
	3	.013	17.355	.04	.41	.01
	4	.008	21.602	.02	.45	.96

Collinearity Diagnostics[a]

Model	Dimension	Variance ... Operational BI utilisation composite
1	1	.00
	2	.03
	3	.83
	4	.14

a. Dependent Variable: Decision-making quality composite

Casewise Diagnostics[a]

Case Number	Std. Residual	Decision-making quality composite	Predicted Value	Residual
90	3.006	4.75	3.3756	1.37436
172	-3.293	2.50	4.0058	-1.50576

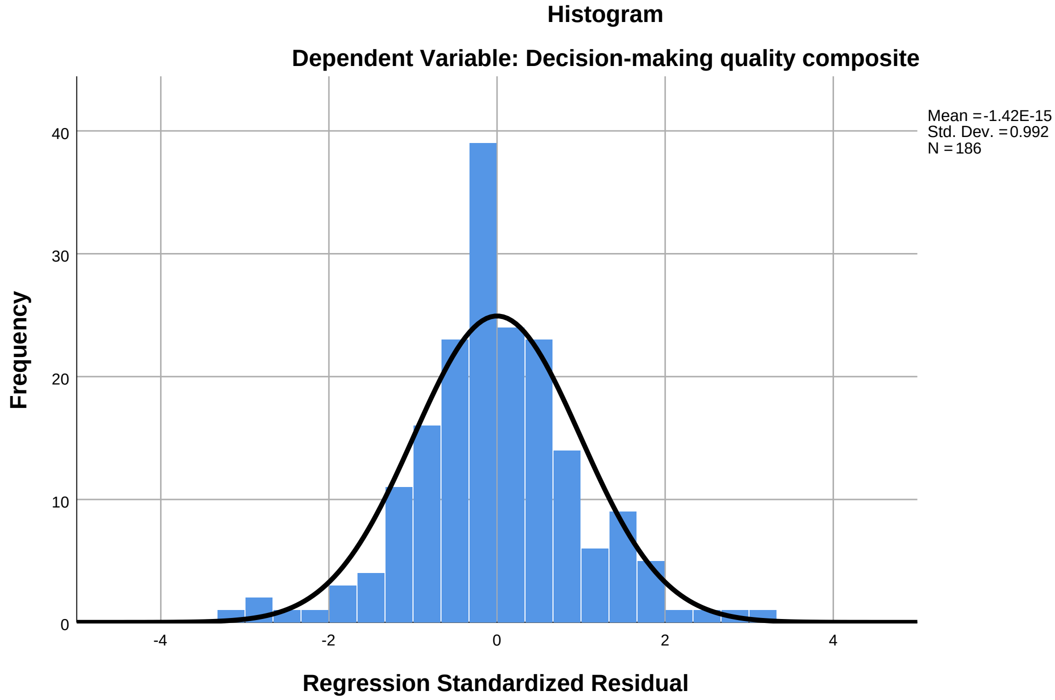
a. Dependent Variable: Decision-making quality composite

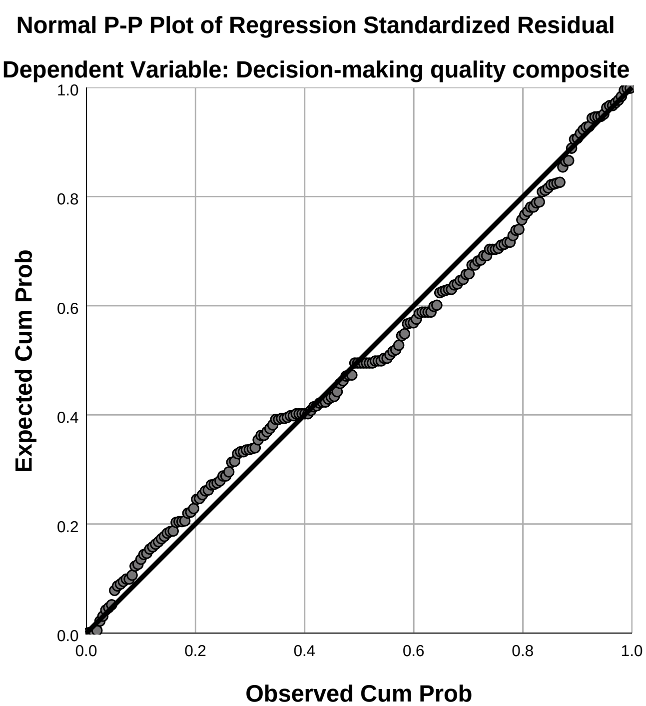
Residuals Statistics[a]

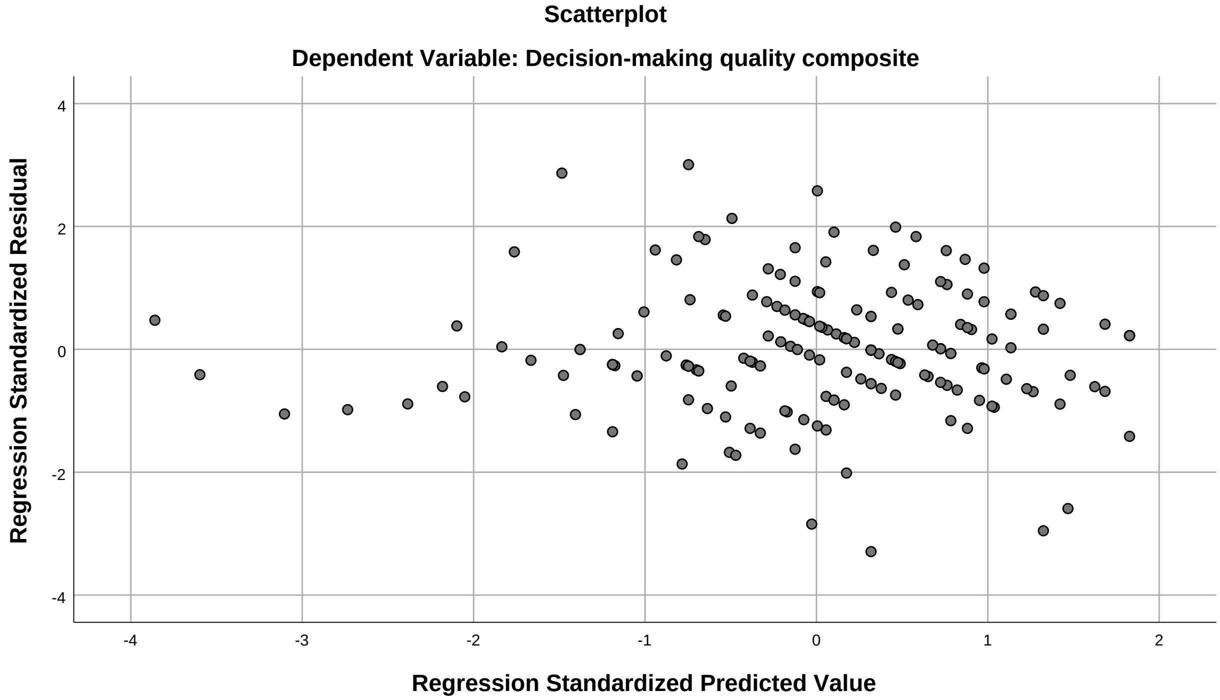
	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	1.5338	4.8980	3.8172	.59154	186
Residual	-1.50576	1.37436	.00000	.45351	186
Std. Predicted Value	-3.860	1.827	.000	1.000	186
Std. Residual	-3.293	3.006	.000	.992	186

a. Dependent Variable: Decision-making quality composite

Charts


Histogram
Dependent Variable: Decision-making quality composite
Mean = -1.42E-15
Std. Dev. = 0.992
N = 186
Frequency
40
30
20
10
0
-4
-2
0
2
4
Regression Standardized Residual


Normal P-P Plot of Regression Standardized Residual
Dependent Variable: Decision-making quality composite
Expected Cum Prob
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Observed Cum Prob




```
 * Model M3: Three BI levels + controls -> DMQ (controlled robustness check).
REGRESSION
  /MISSING LISTWISE
  /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI(95) ZPP
  /CRITERIA=PIN(.05) POUT(.10)
  /NOORIGIN
  /DEPENDENT DMQ_Mean
  /METHOD=ENTER CS_50_99 CS_100_249 JP_SeniorMgr JP_MiddleMgr JP_OpsMgr JP_Oth
er
                IND_Retail IND_Service IND_Technology IND_Other
  /METHOD=ENTER Strategic_Mean Tactical_Mean Operational_Mean
  /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID)
  /CASEWISE PLOT(ZRESID) OUTLIERS(3)
  /SCATTERPLOT=(*ZRESID ,*ZPRED).
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Regression

Notes

Output Created		18-JUN-2026 22:32:48
Comments		
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	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.

Notes

Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI (95) ZPP /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT DMQ_Mean /METHOD=ENTER CS_50_99 CS_100_249 JP_SeniorMgr JP_MiddleMgr JP_OpsMgr JP_Other IND_Retail IND_Service IND_Technology IND_Other /METHOD=ENTER Strategic_Mean Tactical_Mean Operational_Mean /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID) /CASEWISE PLOT (ZRESID) OUTLIERS(3) /SCATTERPLOT=...
Resources	Processor Time	00:00:00.27
	Elapsed Time	00:00:00.32
	Memory Required	7492 bytes
	Additional Memory Required for Residual Plots	816 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Industry: Other (ref: Manufacturing), Job: Middle Manager (ref: CEO), Company size 100-249 (ref: <50), Job: Senior Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Industry: Service (ref: Manufacturing), Company size 50-99 (ref: <50), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)[b]	.	Enter
2	Strategic BI utilisation composite, Operational BI utilisation composite, Tactical BI utilisation composite[b]	.	Enter

a. Dependent Variable: Decision-making quality composite

b. All requested variables entered.

Model Summary[c]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics		
					R Square Change	F Change	df1
1	.350[a]	.123	.073	.71780	.123	2.449	10
2	.805[b]	.648	.622	.45842	.526	85.685	3

Model Summary[c]

Model	Change Statistics		Durbin-Watson
	df2	Sig. F Change	
1	175	.009	
2	172	.000	2.253

a. Predictors: (Constant), Industry: Other (ref: Manufacturing), Job: Middle Manager (ref: CEO), Company size 100-249 (ref: <50), Job: Senior Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Industry: Service (ref: Manufacturing), Company size 50-99 (ref: <50), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)

b. Predictors: (Constant), Industry: Other (ref: Manufacturing), Job: Middle Manager (ref: CEO), Company size 100-249 (ref: <50), Job: Senior Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Industry: Service (ref: Manufacturing), Company size 50-99 (ref: <50), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO), Strategic BI utilisation composite, Operational BI utilisation composite, Tactical BI utilisation composite

c. Dependent Variable: Decision-making quality composite

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	12.619	10	1.262	2.449	.009[b]
	Residual	90.166	175	.515		
	Total	102.785	185			
2	Regression	66.639	13	5.126	24.393	.000[c]
	Residual	36.146	172	.210		
	Total	102.785	185			

a. Dependent Variable: Decision-making quality composite

b. Predictors: (Constant), Industry: Other (ref: Manufacturing), Job: Middle Manager (ref: CEO), Company size 100-249 (ref: <50), Job: Senior Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Industry: Service (ref: Manufacturing), Company size 50-99 (ref: <50), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)

c. Predictors: (Constant), Industry: Other (ref: Manufacturing), Job: Middle Manager (ref: CEO), Company size 100-249 (ref: <50), Job: Senior Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Industry: Service (ref: Manufacturing), Company size 50-99 (ref: <50), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO), Strategic BI utilisation composite, Operational BI utilisation composite, Tactical BI utilisation composite

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	3.832	.261		14.663	.000
	Company size 50-99 (ref: <50)	.321	.132	.210	2.429	.016
	Company size 100-249 (ref: <50)	.145	.143	.087	1.011	.313
	Job: Senior Manager (ref: CEO)	.296	.259	.129	1.142	.255
	Job: Middle Manager (ref: CEO)	.176	.243	.095	.725	.469
	Job: Operational Manager (ref: CEO)	.313	.238	.166	1.314	.190
	Job: Other (ref: CEO)	.130	.233	.087	.559	.577
	Industry: Retail (ref: Manufacturing)	-.588	.231	-.353	-2.548	.012
	Industry: Service (ref: Manufacturing)	-.335	.224	-.203	-1.494	.137
	Industry: Technology (ref: Manufacturing)	-.163	.241	-.082	-.676	.500
	Industry: Other (ref: Manufacturing)	-.402	.241	-.216	-1.672	.096
2	(Constant)	.683	.260		2.627	.009
	Company size 50-99 (ref: <50)	-.016	.088	-.010	-.182	.856
	Company size 100-249 (ref: <50)	-.131	.093	-.079	-1.407	.161
	Job: Senior Manager (ref: CEO)	.097	.167	.042	.578	.564
	Job: Middle Manager (ref: CEO)	-.097	.158	-.052	-.617	.538
	Job: Operational Manager (ref: CEO)	-.067	.156	-.036	-.431	.667
	Job: Other (ref: CEO)	-.022	.151	-.015	-.148	.883
	Industry: Retail (ref: Manufacturing)	-.269	.150	-.161	-1.797	.074
	Industry: Service (ref: Manufacturing)	-.124	.144	-.075	-.857	.392
	Industry: Technology (ref: Manufacturing)	-.169	.154	-.085	-1.097	.274

Coefficients[a]

Model		95.0% Confidence Interval for B		Correlations		
		Lower Bound	Upper Bound	Zero-order	Partial	Part
1	(Constant)	3.316	4.348			
	Company size 50-99 (ref: <50)	.060	.582	.207	.181	.172
	Company size 100-249 (ref: <50)	-.138	.427	-.015	.076	.072
	Job: Senior Manager (ref: CEO)	-.215	.807	.084	.086	.081
	Job: Middle Manager (ref: CEO)	-.303	.655	.046	.055	.051
	Job: Operational Manager (ref: CEO)	-.157	.782	.120	.099	.093
	Job: Other (ref: CEO)	-.330	.591	-.168	.042	.040
	Industry: Retail (ref: Manufacturing)	-1.043	-.132	-.197	-.189	-.180
	Industry: Service (ref: Manufacturing)	-.777	.107	.015	-.112	-.106
	Industry: Technology (ref: Manufacturing)	-.638	.312	.163	-.051	-.048
	Industry: Other (ref: Manufacturing)	-.877	.073	-.031	-.125	-.118
2	(Constant)	.170	1.196			
	Company size 50-99 (ref: <50)	-.189	.157	.207	-.014	-.008
	Company size 100-249 (ref: <50)	-.316	.053	-.015	-.107	-.064
	Job: Senior Manager (ref: CEO)	-.233	.426	.084	.044	.026
	Job: Middle Manager (ref: CEO)	-.409	.214	.046	-.047	-.028
	Job: Operational Manager (ref: CEO)	-.375	.240	.120	-.033	-.019
	Job: Other (ref: CEO)	-.321	.277	-.168	-.011	-.007
	Industry: Retail (ref: Manufacturing)	-.564	.026	-.197	-.136	-.081
	Industry: Service (ref: Manufacturing)	-.408	.161	.015	-.065	-.039
	Industry: Technology (ref: Manufacturing)	-.474	.135	.163	-.083	-.050

Coefficients[a]

Model		Collinearity Statistics Tolerance	Collinearity Statistics VIF
1	(Constant)		
	Company size 50-99 (ref: <50)	.669	1.495
	Company size 100-249 (ref: <50)	.681	1.467
	Job: Senior Manager (ref: CEO)	.396	2.525
	Job: Middle Manager (ref: CEO)	.295	3.390
	Job: Operational Manager (ref: CEO)	.314	3.188
	Job: Other (ref: CEO)	.209	4.779
	Industry: Retail (ref: Manufacturing)	.262	3.822
	Industry: Service (ref: Manufacturing)	.270	3.697
	Industry: Technology (ref: Manufacturing)	.344	2.907
	Industry: Other (ref: Manufacturing)	.300	3.330
2	(Constant)		
	Company size 50-99 (ref: <50)	.621	1.610
	Company size 100-249 (ref: <50)	.651	1.537
	Job: Senior Manager (ref: CEO)	.389	2.573
	Job: Middle Manager (ref: CEO)	.285	3.507
	Job: Operational Manager (ref: CEO)	.298	3.354
	Job: Other (ref: CEO)	.202	4.943
	Industry: Retail (ref: Manufacturing)	.254	3.940
	Industry: Service (ref: Manufacturing)	.267	3.741
	Industry: Technology (ref: Manufacturing)	.342	2.925

Coefficients[a]

Model	Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
Industry: Other (ref: Manufacturing)	-.195	.154	-.105	-1.263	.208
Strategic BI utilisation composite	.261	.071	.271	3.684	.000
Tactical BI utilisation composite	.298	.082	.282	3.621	.000
Operational BI utilisation composite	.334	.071	.333	4.721	.000

Coefficients[a]

Model	95.0% Confidence Interval for B Lower Bound	95.0% Confidence Interval for B Upper Bound	Correlations Zero-order	Correlations Partial	Correlations Part
Industry: Other (ref: Manufacturing)	-.499	.110	-.031	-.096	-.057
Strategic BI utilisation composite	.121	.401	.698	.270	.167
Tactical BI utilisation composite	.135	.460	.714	.266	.164
Operational BI utilisation composite	.194	.474	.717	.339	.213

Coefficients[a]

Model	Collinearity Statistics Tolerance	Collinearity Statistics VIF
Industry: Other (ref: Manufacturing)	.298	3.358
Strategic BI utilisation composite	.377	2.651
Tactical BI utilisation composite	.337	2.966
Operational BI utilisation composite	.410	2.437

a. Dependent Variable: Decision-making quality composite

Excluded Variables[a]

Model		Beta In	t	Sig.	Partial Correlation	Collinearity . Tolerance
1	Strategic BI utilisation composite	.677[b]	12.241	.000	.680	.886
	Tactical BI utilisation composite	.697[b]	12.853	.000	.698	.880
	Operational BI utilisation composite	.710[b]	12.210	.000	.679	.803

Excluded Variables[a]

Model		Collinearity Statistics VIF	Minimum Tolerance
1	Strategic BI utilisation composite	1.129	.208
	Tactical BI utilisation composite	1.136	.206
	Operational BI utilisation composite	1.245	.209

a. Dependent Variable: Decision-making quality composite

b. Predictors in the Model: (Constant), Industry: Other (ref: Manufacturing), Job: Middle Manager (ref: CEO), Company size 100-249 (ref: <50), Job: Senior Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Industry: Service (ref: Manufacturing), Company size 50-99 (ref: <50), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Company size 50-99 (ref: <50)	Company size 100-249 (ref: <50)
1	1	3.638	1.000	.00	.01	.01
	2	1.515	1.550	.00	.05	.06
	3	1.218	1.728	.00	.00	.00
	4	1.037	1.873	.00	.00	.00
	5	.978	1.929	.00	.00	.00
	6	.954	1.953	.00	.04	.08
	7	.774	2.168	.00	.00	.01
	8	.558	2.553	.00	.09	.23
	9	.246	3.843	.01	.81	.54
	10	.055	8.165	.00	.00	.05
	11	.028	11.479	.99	.00	.01
2	1	6.536	1.000	.00	.00	.00
	2	1.516	2.076	.00	.04	.06
	3	1.219	2.316	.00	.00	.00
	4	1.037	2.510	.00	.00	.00
	5	.978	2.585	.00	.00	.00
	6	.954	2.618	.00	.04	.08
	7	.774	2.906	.00	.00	.01
	8	.559	3.421	.00	.08	.22
	9	.262	4.999	.00	.79	.56
	10	.079	9.125	.00	.00	.02
	11	.053	11.087	.00	.00	.03
	12	.016	20.499	.60	.00	.00
	13	.011	24.382	.39	.03	.02
	14	.008	28.651	.01	.01	.00

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions Job: Senior Manager (ref: CEO)	Job: Middle Manager (ref: CEO)	Job: Operational Manager (ref: CEO)	Job: Other (ref: CEO)	Industry: Retail (ref: Manufacturing)
1	1	.00	.00	.00	.00	.00
	2	.00	.00	.04	.02	.00
	3	.01	.06	.01	.01	.03
	4	.11	.02	.00	.00	.04
	5	.16	.03	.01	.00	.03
	6	.01	.01	.00	.00	.01
	7	.01	.07	.07	.00	.03
	8	.01	.01	.12	.05	.00
	9	.02	.03	.01	.00	.01
	10	.34	.40	.41	.60	.53
	11	.33	.38	.33	.31	.32
2	1	.00	.00	.00	.00	.00
	2	.00	.00	.04	.02	.00
	3	.01	.06	.01	.01	.03
	4	.10	.01	.00	.00	.04
	5	.16	.03	.01	.00	.03
	6	.01	.01	.00	.00	.01
	7	.01	.07	.06	.00	.03
	8	.01	.01	.11	.05	.00
	9	.01	.01	.00	.00	.00
	10	.09	.11	.10	.04	.32
	11	.51	.59	.59	.78	.30
	12	.04	.03	.03	.03	.13
	13	.03	.04	.01	.04	.11
	14	.02	.03	.04	.03	.00

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions: Industry: Service (ref: Manufacturing)	Industry: Technology (ref: Manufacturing)	Industry: Other (ref: Manufacturing)	Strategic BI utilisation composite	Tactical BI utilisation composite
1	1	.00	.00	.00		
	2	.00	.02	.01		
	3	.04	.00	.00		
	4	.01	.04	.05		
	5	.01	.03	.02		
	6	.01	.09	.08		
	7	.06	.03	.01		
	8	.01	.02	.01		
	9	.01	.07	.03		
	10	.55	.35	.51		
	11	.29	.34	.26		
2	1	.00	.00	.00	.00	.00
	2	.00	.02	.01	.00	.00
	3	.04	.00	.00	.00	.00
	4	.01	.04	.05	.00	.00
	5	.01	.03	.02	.00	.00
	6	.01	.09	.08	.00	.00
	7	.06	.03	.01	.00	.00
	8	.01	.02	.01	.00	.00
	9	.00	.04	.02	.00	.00
	10	.33	.43	.33	.03	.01
	11	.32	.15	.30	.00	.00
	12	.15	.09	.12	.25	.04
	13	.04	.06	.05	.27	.00
	14	.00	.00	.00	.44	.94

Collinearity Diagnostics[a]

Model	Dimension	Variance ... Operational BI utilisation composite
1	1	
	2	
	3	
	4	
	5	
	6	
	7	
	8	
	9	
	10	
	11	
2	1	.00
	2	.00
	3	.00
	4	.00
	5	.00
	6	.00
	7	.00
	8	.00
	9	.00
	10	.02
	11	.00
	12	.07
	13	.71
	14	.19

a. Dependent Variable: Decision-making quality composite

Casewise Diagnostics[a]

Case Number	Std. Residual	Decision-making quality composite	Predicted Value	Residual
90	3.066	4.75	3.3446	1.40543

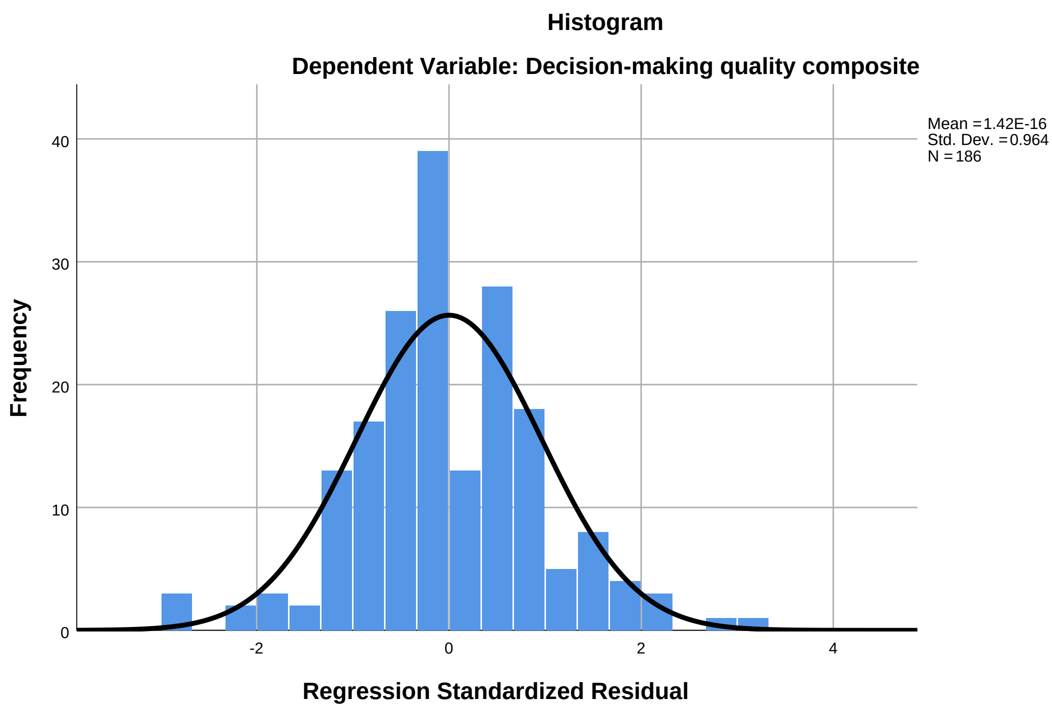
a. Dependent Variable: Decision-making quality composite

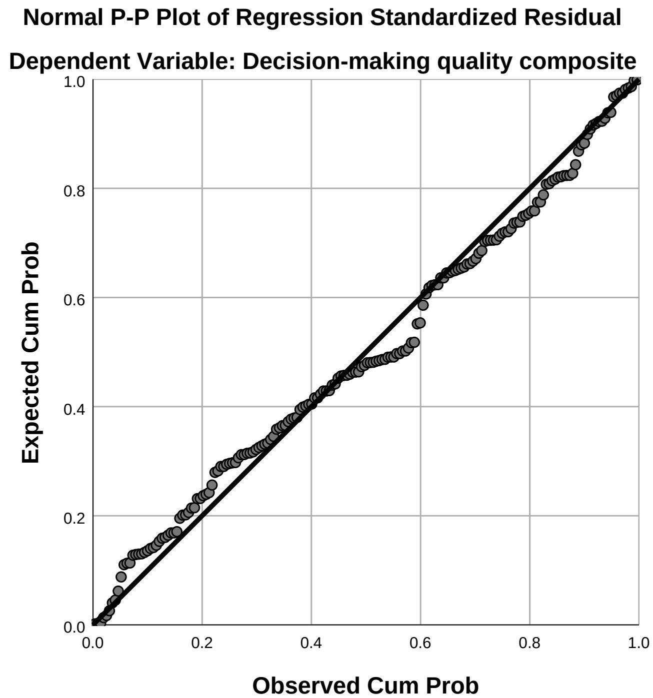
Residuals Statistics[a]

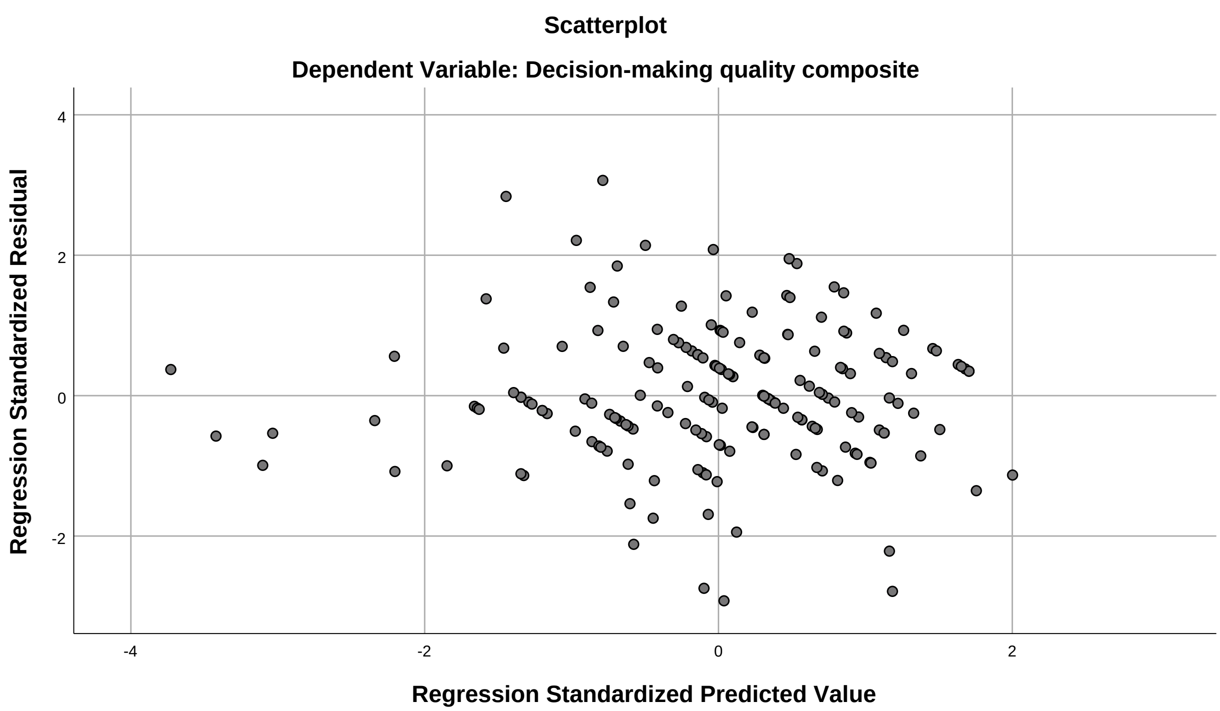
	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	1.5795	5.0186	3.8172	.60018	186
Residual	-1.33969	1.40543	.00000	.44202	186
Std. Predicted Value	-3.728	2.002	.000	1.000	186
Std. Residual	-2.922	3.066	.000	.964	186

a. Dependent Variable: Decision-making quality composite

Charts




Normal P-P Plot of Regression Standardized Residual
Dependent Variable: Decision-making quality composite
Expected Cum Prob
Observed Cum Prob
0.0
0.2
0.4
0.6
0.8
1.0


Scatterplot
Dependent Variable: Decision-making quality composite
Regression Standardized Residual
Regression Standardized Predicted Value
4
2
0
-2
-4

```
* -------------------- 10. ROBUSTNESS: ALTERNATIVE SPECIFICATIONS ------------
--------

* M4: Overall BI (single composite) -> DMQ.
REGRESSION
  /MISSING LISTWISE
  /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI(95)
  /CRITERIA=PIN(.05) POUT(.10)
  /NOORIGIN
  /DEPENDENT DMQ_Mean
  /METHOD=ENTER CS_50_99 CS_100_249 JP_SeniorMgr JP_MiddleMgr JP_OpsMgr JP_Oth
er
                IND_Retail IND_Service IND_Technology IND_Other Overall_BI_Mea
n
  /RESIDUALS DURBIN.
```

Regression

Notes

Output Created		18-JUN-2026 22:32:48
Comments		
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	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.

Notes

Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI (95) /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT DMQ_Mean /METHOD=ENTER CS_50_99 CS_100_249 JP_SeniorMgr JP_MiddleMgr JP_OpsMgr JP_Other IND_Retail IND_Service IND_Technology IND_Other Overall_BI_Mean...
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Memory Required	6236 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Overall BI utilisation (all 9 items), Job: Senior Manager (ref: CEO), Industry: Other (ref: Manufacturing), Company size 100-249 (ref: <50), Job: Middle Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Industry: Service (ref: Manufacturing), Job: Operational Manager (ref: CEO), Company size 50-99 (ref: <50), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)[b]	.	Enter

a. Dependent Variable: Decision-making quality composite

b. All requested variables entered.

Model Summary[b]

					Change Statistics		
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	F Change	df1
1	.805[a]	.647	.625	.45634	.647	29.053	11

Model Summary[b]

	Change Statistics		
Model	df2	Sig. F Change	Durbin-Watson
1	174	.000	2.254

a. Predictors: (Constant), Overall BI utilisation (all 9 items), Job: Senior Manager (ref: CEO), Industry: Other (ref: Manufacturing), Company size 100-249 (ref: <50), Job: Middle Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Industry: Service (ref: Manufacturing), Job: Operational Manager (ref: CEO), Company size 50-99 (ref: <50), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)

b. Dependent Variable: Decision-making quality composite

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	66.551	11	6.050	29.053	.000[b]
	Residual	36.234	174	.208		
	Total	102.785	185			

a. Dependent Variable: Decision-making quality composite

b. Predictors: (Constant), Overall BI utilisation (all 9 items), Job: Senior Manager (ref: CEO), Industry: Other (ref: Manufacturing), Company size 100-249 (ref: <50), Job: Middle Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Industry: Service (ref: Manufacturing), Job: Operational Manager (ref: CEO), Company size 50-99 (ref: <50), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	.708	.256		2.771	.006
	Company size 50-99 (ref: <50)	-.011	.086	-.007	-.122	.903
	Company size 100-249 (ref: <50)	-.125	.092	-.075	-1.348	.179
	Job: Senior Manager (ref: CEO)	.094	.165	.041	.570	.570
	Job: Middle Manager (ref: CEO)	-.103	.155	-.055	-.664	.508
	Job: Operational Manager (ref: CEO)	-.064	.153	-.034	-.417	.677
	Job: Other (ref: CEO)	-.030	.149	-.020	-.202	.840
	Industry: Retail (ref: Manufacturing)	-.279	.148	-.168	-1.888	.061
	Industry: Service (ref: Manufacturing)	-.121	.143	-.074	-.846	.398
	Industry: Technology (ref: Manufacturing)	-.177	.153	-.089	-1.153	.250
	Industry: Other (ref: Manufacturing)	-.198	.153	-.106	-1.288	.200
	Overall BI utilisation (all 9 items)	.888	.055	.791	16.093	.000

Coefficients[a]

Model		95.0% Confidence Interval for B		Collinearity Statistics	
		Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	.204	1.212		
	Company size 50-99 (ref: <50)	-.181	.160	.631	1.585
	Company size 100-249 (ref: <50)	-.307	.058	.659	1.517
	Job: Senior Manager (ref: CEO)	-.232	.420	.394	2.539
	Job: Middle Manager (ref: CEO)	-.410	.203	.291	3.433
	Job: Operational Manager (ref: CEO)	-.366	.238	.306	3.264
	Job: Other (ref: CEO)	-.323	.263	.208	4.801
	Industry: Retail (ref: Manufacturing)	-.571	.013	.257	3.888
	Industry: Service (ref: Manufacturing)	-.404	.161	.268	3.729
	Industry: Technology (ref: Manufacturing)	-.479	.126	.344	2.907
	Industry: Other (ref: Manufacturing)	-.501	.105	.298	3.353
	Overall BI utilisation (all 9 items)	.779	.997	.838	1.193

a. Dependent Variable: Decision-making quality composite

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Company size 50-99 (ref: <50)	Company size 100-249 (ref: <50)
1	1	4.600	1.000	.00	.01	.01
	2	1.515	1.742	.00	.04	.06
	3	1.218	1.943	.00	.00	.00
	4	1.037	2.106	.00	.00	.00
	5	.978	2.169	.00	.00	.00
	6	.954	2.196	.00	.04	.08
	7	.774	2.438	.00	.00	.01
	8	.558	2.870	.00	.08	.22
	9	.253	4.268	.00	.78	.54
	10	.057	9.020	.00	.00	.06
	11	.046	10.016	.04	.01	.00
	12	.011	20.301	.95	.03	.01

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions Job: Senior Manager (ref: CEO)	Job: Middle Manager (ref: CEO)	Job: Operational Manager (ref: CEO)	Job: Other (ref: CEO)	Industry: Retail (ref: Manufacturing)
1	1	.00	.00	.00	.00	.00
	2	.00	.00	.04	.02	.00
	3	.01	.06	.01	.01	.03
	4	.10	.01	.00	.00	.04
	5	.16	.03	.01	.00	.03
	6	.01	.01	.00	.00	.01
	7	.01	.07	.06	.00	.03
	8	.01	.01	.11	.05	.00
	9	.02	.03	.01	.00	.01
	10	.12	.14	.15	.29	.67
	11	.52	.63	.59	.60	.02
	12	.03	.03	.01	.03	.16

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions: Industry: Service (ref: Manufacturing)	Industry: Technology (ref: Manufacturing)	Industry: Other (ref: Manufacturing)	Overall BI utilisation (all 9 items)
1	1	.00	.00	.00	.00
	2	.00	.02	.01	.00
	3	.04	.00	.00	.00
	4	.01	.04	.05	.00
	5	.01	.03	.02	.00
	6	.01	.09	.08	.00
	7	.06	.03	.01	.00
	8	.01	.02	.01	.00
	9	.00	.05	.03	.00
	10	.70	.54	.65	.04
	11	.02	.09	.02	.18
	12	.13	.08	.11	.78

a. Dependent Variable: Decision-making quality composite

Residuals Statistics[a]

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	1.5962	5.0137	3.8172	.59978	186
Residual	-1.33668	1.38975	.00000	.44256	186
Std. Predicted Value	-3.703	1.995	.000	1.000	186
Std. Residual	-2.929	3.045	.000	.970	186

a. Dependent Variable: Decision-making quality composite

```
 * M5: Managerial BI + Operational BI -> DMQ.
REGRESSION
  /MISSING LISTWISE
  /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI(95)
  /CRITERIA=PIN(.05) POUT(.10)
  /NOORIGIN
```

```
  /DEPENDENT DMQ_Mean
  /METHOD=ENTER CS_50_99 CS_100_249 JP_SeniorMgr JP_MiddleMgr JP_OpsMgr JP_Oth
er
                IND_Retail IND_Service IND_Technology IND_Other
                Managerial_BI_MeanOperational_Mean
  /RESIDUALS DURBIN.
```

Regression

Notes

Output Created		18-JUN-2026 22:32:48
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.

Notes

Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI (95) /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT DMQ_Mean /METHOD=ENTER CS_50_99 CS_100_249 JP_SeniorMgr JP_MiddleMgr JP_OpsMgr JP_Other IND_Retail IND_Service IND_Technology IND_Other Managerial_BI_Mean Operational_Mean...
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.02
	Memory Required	6820 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Operational BI utilisation composite, Industry: Service (ref: Manufacturing), Company size 100-249 (ref: <50), Job: Senior Manager (ref: CEO), Job: Middle Manager (ref: CEO), Industry: Other (ref: Manufacturing), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Company size 50-99 (ref: <50), Managerial BI (Strategic + Tactical), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)[b]	.	Enter

a. Dependent Variable: Decision-making quality composite

b. All requested variables entered.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics		
					R Square Change	F Change	df1
1	.805[a]	.648	.624	.45719	.648	26.562	12

Model Summary[b]

Model	Change Statistics		Durbin-Watson
	df2	Sig. F Change	
1	173	.000	2.250

a. Predictors: (Constant), Operational BI utilisation composite, Industry: Service (ref: Manufacturing), Company size 100-249 (ref: <50), Job: Senior Manager (ref: CEO), Job: Middle Manager (ref: CEO), Industry: Other (ref: Manufacturing), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Company size 50-99 (ref: <50), Managerial BI (Strategic + Tactical), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)

b. Dependent Variable: Decision-making quality composite

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	66.624	12	5.552	26.562	.000[b]
	Residual	36.161	173	.209		
	Total	102.785	185			

a. Dependent Variable: Decision-making quality composite

b. Predictors: (Constant), Operational BI utilisation composite, Industry: Service (ref: Manufacturing), Company size 100-249 (ref: <50), Job: Senior Manager (ref: CEO), Job: Middle Manager (ref: CEO), Industry: Other (ref: Manufacturing), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Company size 50-99 (ref: <50), Managerial BI (Strategic + Tactical), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	.684	.259		2.640	.009
	Company size 50-99 (ref: <50)	-.017	.087	-.011	-.192	.848
	Company size 100-249 (ref: <50)	-.130	.093	-.078	-1.401	.163
	Job: Senior Manager (ref: CEO)	.101	.166	.044	.607	.545
	Job: Middle Manager (ref: CEO)	-.093	.157	-.050	-.595	.553
	Job: Operational Manager (ref: CEO)	-.060	.153	-.032	-.394	.694
	Job: Other (ref: CEO)	-.018	.150	-.012	-.120	.904
	Industry: Retail (ref: Manufacturing)	-.269	.149	-.161	-1.803	.073
	Industry: Service (ref: Manufacturing)	-.121	.143	-.074	-.846	.399
	Industry: Technology (ref: Manufacturing)	-.171	.154	-.086	-1.116	.266
	Industry: Other (ref: Manufacturing)	-.194	.154	-.104	-1.262	.209
	Managerial BI (Strategic + Tactical)	.556	.072	.516	7.702	.000
	Operational BI utilisation composite	.336	.070	.335	4.791	.000

Coefficients[a]

Model		95.0% Confidence Interval for B Lower Bound	Upper Bound	Collinearity Statistics Tolerance	VIF
1	(Constant)	.173	1.196		
	Company size 50-99 (ref: <50)	-.189	.156	.622	1.608
	Company size 100-249 (ref: <50)	-.314	.053	.652	1.534
	Job: Senior Manager (ref: CEO)	-.227	.428	.392	2.551
	Job: Middle Manager (ref: CEO)	-.402	.216	.288	3.473
	Job: Operational Manager (ref: CEO)	-.363	.242	.306	3.269
	Job: Other (ref: CEO)	-.315	.278	.205	4.889
	Industry: Retail (ref: Manufacturing)	-.563	.025	.254	3.940
	Industry: Service (ref: Manufacturing)	-.404	.162	.268	3.729
	Industry: Technology (ref: Manufacturing)	-.475	.132	.343	2.917
	Industry: Other (ref: Manufacturing)	-.498	.110	.298	3.357
	Managerial BI (Strategic + Tactical)	.413	.698	.453	2.210
	Operational BI utilisation composite	.198	.475	.415	2.409

a. Dependent Variable: Decision-making quality composite

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Company size 50-99 (ref: <50)	Company size 100-249 (ref: <50)
1	1	5.566	1.000	.00	.01	.00
	2	1.516	1.916	.00	.04	.06
	3	1.219	2.137	.00	.00	.00
	4	1.037	2.317	.00	.00	.00
	5	.978	2.386	.00	.00	.00
	6	.954	2.416	.00	.04	.08
	7	.774	2.682	.00	.00	.01
	8	.558	3.157	.00	.08	.22
	9	.257	4.654	.00	.79	.55
	10	.068	9.077	.00	.01	.04
	11	.053	10.283	.00	.00	.02
	12	.013	20.647	.97	.02	.00
	13	.009	25.268	.02	.01	.01

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions: Job: Senior Manager (ref: CEO)	Job: Middle Manager (ref: CEO)	Job: Operational Manager (ref: CEO)	Job: Other (ref: CEO)	Industry: Retail (ref: Manufacturing)
1	1	.00	.00	.00	.00	.00
	2	.00	.00	.04	.02	.00
	3	.01	.06	.01	.01	.03
	4	.10	.01	.00	.00	.04
	5	.16	.03	.01	.00	.03
	6	.01	.01	.00	.00	.01
	7	.01	.07	.06	.00	.03
	8	.01	.01	.11	.05	.00
	9	.01	.02	.00	.00	.00
	10	.05	.06	.05	.01	.42
	11	.56	.66	.67	.82	.22
	12	.06	.05	.03	.05	.21
	13	.01	.03	.01	.03	.02

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions: Industry: Service (ref: Manufacturing)	Industry: Technology (ref: Manufacturing)	Industry: Other (ref: Manufacturing)	Managerial BI (Strategic + Tactical)	Operational BI utilisation composite
1	1	.00	.00	.00	.00	.00
	2	.00	.02	.01	.00	.00
	3	.04	.00	.00	.00	.00
	4	.01	.04	.05	.00	.00
	5	.01	.03	.02	.00	.00
	6	.01	.09	.08	.00	.00
	7	.06	.03	.01	.00	.00
	8	.01	.02	.01	.00	.00
	9	.00	.04	.02	.00	.00
	10	.44	.50	.42	.04	.05
	11	.24	.09	.22	.00	.01
	12	.17	.12	.15	.16	.05
	13	.00	.01	.00	.79	.89

a. Dependent Variable: Decision-making quality composite

Residuals Statistics[a]

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	1.5865	5.0132	3.8172	.60011	186
Residual	-1.33450	1.39963	.00000	.44211	186
Std. Predicted Value	-3.717	1.993	.000	1.000	186
Std. Residual	-2.919	3.061	.000	.967	186

a. Dependent Variable: Decision-making quality composite

```
* -------------------- 11. SENSITIVITY: EXCLUDING STRAIGHT-LINERS ------------
--------

* Re-run primary model excluding cases with SD = 0 across 13 BI/DMQ items.
```

```
TEMPORARY.
SELECT IF (BI_DMQ_SD > 0).
REGRESSION
  /MISSING LISTWISE
  /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI(95)
  /CRITERIA=PIN(.05) POUT(.10)
  /NOORIGIN
  /DEPENDENT DMQ_Mean
  /METHOD=ENTER CS_50_99 CS_100_249 JP_SeniorMgr JP_MiddleMgr JP_OpsMgr JP_Oth
er
                IND_Retail IND_Service IND_Technology IND_Other
  /METHOD=ENTER Strategic_Mean Tactical_Mean Operational_Mean.
```

Regression

Notes

Output Created		18-JUN-2026 22:32:48
Comments		
Input	Data	C: \Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	171
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.

Notes

Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI (95) /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT DMQ_Mean /METHOD=ENTER CS_50_99 CS_100_249 JP_SeniorMgr JP_MiddleMgr JP_OpsMgr JP_Other IND_Retail IND_Service IND_Technology IND_Other /METHOD=ENTER Strategic_Mean Tactical_Mean ...
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01
	Memory Required	7444 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Industry: Other (ref: Manufacturing), Company size 50-99 (ref: <50), Job: Senior Manager (ref: CEO), Job: Middle Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Company size 100-249 (ref: <50), Industry: Service (ref: Manufacturing), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)[b]	.	Enter
2	Strategic BI utilisation composite, Operational BI utilisation composite, Tactical BI utilisation composite[b]	.	Enter

a. Dependent Variable: Decision-making quality composite

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics		
					R Square Change	F Change	df1
1	.360[a]	.129	.075	.71610	.129	2.378	10
2	.788[b]	.621	.589	.47717	.491	67.781	3

Model Summary

Model	Change Statistics	
	df2	Sig. F Change
1	160	.012
2	157	.000

a. Predictors: (Constant), Industry: Other (ref: Manufacturing), Company size 50-99 (ref: <50), Job: Senior Manager (ref: CEO), Job: Middle Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Company size 100-249 (ref: <50), Industry: Service (ref: Manufacturing), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)

b. Predictors: (Constant), Industry: Other (ref: Manufacturing), Company size 50-99 (ref: <50), Job: Senior Manager (ref: CEO), Job: Middle Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Company size 100-249 (ref: <50), Industry: Service (ref: Manufacturing), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO), Strategic BI utilisation composite, Operational

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	12.193	10	1.219	2.378	.012[b]
	Residual	82.047	160	.513		
	Total	94.240	170			
2	Regression	58.492	13	4.499	19.761	.000[c]
	Residual	35.748	157	.228		
	Total	94.240	170			

a. Dependent Variable: Decision-making quality composite

b. Predictors: (Constant), Industry: Other (ref: Manufacturing), Company size 50-99 (ref: <50), Job: Senior Manager (ref: CEO), Job: Middle Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Company size 100-249 (ref: <50), Industry: Service (ref: Manufacturing), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)

c. Predictors: (Constant), Industry: Other (ref: Manufacturing), Company size 50-99 (ref: <50), Job: Senior Manager (ref: CEO), Job: Middle Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Company size 100-249 (ref: <50), Industry: Service (ref: Manufacturing), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO), Strategic BI utilisation composite, Operational BI utilisation composite, Tactical BI utilisation composite

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	3.869	.265		14.576	.000
	Company size 50-99 (ref: <50)	.364	.138	.233	2.640	.009
	Company size 100-249 (ref: <50)	.138	.144	.084	.960	.338
	Job: Senior Manager (ref: CEO)	.323	.266	.137	1.216	.226
	Job: Middle Manager (ref: CEO)	.129	.247	.068	.521	.603
	Job: Operational Manager (ref: CEO)	.353	.242	.181	1.458	.147
	Job: Other (ref: CEO)	.146	.235	.098	.624	.533
	Industry: Retail (ref: Manufacturing)	-.607	.241	-.363	-2.524	.013
	Industry: Service (ref: Manufacturing)	-.379	.232	-.234	-1.632	.105
	Industry: Technology (ref: Manufacturing)	-.257	.254	-.126	-1.011	.314
	Industry: Other (ref: Manufacturing)	-.523	.251	-.281	-2.081	.039
2	(Constant)	.761	.283		2.685	.008
	Company size 50-99 (ref: <50)	.002	.096	.001	.021	.983
	Company size 100-249 (ref: <50)	-.136	.099	-.083	-1.377	.170
	Job: Senior Manager (ref: CEO)	.136	.179	.057	.761	.448
	Job: Middle Manager (ref: CEO)	-.113	.167	-.060	-.676	.500
	Job: Operational Manager (ref: CEO)	-.064	.166	-.033	-.383	.702
	Job: Other (ref: CEO)	-.013	.159	-.009	-.083	.934
	Industry: Retail (ref: Manufacturing)	-.315	.163	-.188	-1.935	.055
	Industry: Service (ref: Manufacturing)	-.154	.156	-.095	-.986	.326
	Industry: Technology (ref: Manufacturing)	-.220	.170	-.108	-1.294	.198

Coefficients[a]

Model		95.0% Confidence Interval for B Lower Bound	Upper Bound	Collinearity Statistics Tolerance	VIF
1	(Constant)	3.345	4.393		
	Company size 50-99 (ref: <50)	.092	.636	.698	1.433
	Company size 100-249 (ref: <50)	-.146	.423	.706	1.417
	Job: Senior Manager (ref: CEO)	-.202	.848	.430	2.326
	Job: Middle Manager (ref: CEO)	-.359	.616	.316	3.165
	Job: Operational Manager (ref: CEO)	-.125	.832	.353	2.833
	Job: Other (ref: CEO)	-.317	.610	.221	4.531
	Industry: Retail (ref: Manufacturing)	-1.082	-.132	.263	3.796
	Industry: Service (ref: Manufacturing)	-.838	.080	.265	3.768
	Industry: Technology (ref: Manufacturing)	-.758	.245	.350	2.861
	Industry: Other (ref: Manufacturing)	-1.020	-.027	.298	3.360
2	(Constant)	.201	1.320		
	Company size 50-99 (ref: <50)	-.188	.192	.638	1.568
	Company size 100-249 (ref: <50)	-.330	.059	.671	1.491
	Job: Senior Manager (ref: CEO)	-.217	.488	.423	2.364
	Job: Middle Manager (ref: CEO)	-.443	.217	.306	3.266
	Job: Operational Manager (ref: CEO)	-.392	.265	.333	3.006
	Job: Other (ref: CEO)	-.327	.301	.213	4.685
	Industry: Retail (ref: Manufacturing)	-.636	.007	.256	3.912
	Industry: Service (ref: Manufacturing)	-.462	.154	.262	3.820
	Industry: Technology (ref: Manufacturing)	-.555	.116	.347	2.883

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
	Industry: Other (ref: Manufacturing)	-.240	.169	-.129	-1.419	.158
	Strategic BI utilisation composite	.261	.074	.271	3.525	.001
	Tactical BI utilisation composite	.294	.086	.277	3.411	.001
	Operational BI utilisation composite	.325	.075	.324	4.357	.000

Coefficients[a]

Model		95.0% Confidence Interval for B Lower Bound	95.0% Confidence Interval for B Upper Bound	Collinearity Statistics Tolerance	Collinearity Statistics VIF
	Industry: Other (ref: Manufacturing)	-.573	.094	.293	3.414
	Strategic BI utilisation composite	.115	.408	.408	2.450
	Tactical BI utilisation composite	.124	.464	.368	2.720
	Operational BI utilisation composite	.178	.472	.437	2.287

a. Dependent Variable: Decision-making quality composite

Excluded Variables[a]

Model		Beta In	t	Sig.	Partial Correlation	Collinearity . Tolerance
1	Strategic BI utilisation composite	.648[b]	10.901	.000	.654	.886
	Tactical BI utilisation composite	.668[b]	11.355	.000	.669	.875
	Operational BI utilisation composite	.680[b]	10.721	.000	.648	.790

Excluded Variables[a]

Model		VIF	Minimum Tolerance
		Collinearity Statistics	
1	Strategic BI utilisation composite	1.128	.220
	Tactical BI utilisation composite	1.143	.217
	Operational BI utilisation composite	1.266	.221

a. Dependent Variable: Decision-making quality composite

b. Predictors in the Model: (Constant), Industry: Other (ref: Manufacturing), Company size 50-99 (ref: <50), Job: Senior Manager (ref: CEO), Job: Middle Manager (ref: CEO), Industry: Technology (ref: Manufacturing), Job: Operational Manager (ref: CEO), Company size 100-249 (ref: <50), Industry: Service (ref: Manufacturing), Industry: Retail (ref: Manufacturing), Job: Other (ref: CEO)

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Company size 50-99 (ref: <50)	Company size 100-249 (ref: <50)
1	1	3.608	1.000	.00	.01	.01
	2	1.519	1.541	.00	.05	.06
	3	1.241	1.705	.00	.00	.00
	4	1.054	1.850	.00	.01	.00
	5	.971	1.928	.00	.00	.01
	6	.887	2.017	.00	.05	.09
	7	.773	2.160	.00	.00	.01
	8	.593	2.467	.00	.10	.20
	9	.270	3.659	.01	.77	.56
	10	.056	8.029	.00	.00	.05
	11	.029	11.165	.99	.00	.01
2	1	6.501	1.000	.00	.00	.00
	2	1.521	2.067	.00	.04	.06
	3	1.241	2.289	.00	.00	.00
	4	1.054	2.483	.00	.01	.00
	5	.972	2.587	.00	.00	.01
	6	.888	2.707	.00	.05	.08

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions: Job: Senior Manager (ref: CEO)	Job: Middle Manager (ref: CEO)	Job: Operational Manager (ref: CEO)	Job: Other (ref: CEO)	Industry: Retail (ref: Manufacturing)
1	1	.00	.00	.00	.00	.00
	2	.00	.00	.04	.01	.00
	3	.00	.07	.02	.01	.02
	4	.13	.01	.00	.00	.04
	5	.15	.02	.02	.00	.05
	6	.03	.01	.00	.00	.01
	7	.00	.07	.10	.00	.02
	8	.01	.02	.11	.04	.00
	9	.02	.04	.00	.01	.01
	10	.38	.42	.43	.66	.48
	11	.25	.33	.28	.25	.37
2	1	.00	.00	.00	.00	.00
	2	.00	.00	.04	.02	.00
	3	.00	.07	.01	.01	.02
	4	.13	.01	.00	.00	.04
	5	.15	.02	.01	.00	.04
	6	.03	.01	.00	.00	.01

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions: Industry: Service (ref: Manufacturing)	Industry: Technology (ref: Manufacturing)	Industry: Other (ref: Manufacturing)	Strategic BI utilisation composite	Tactical BI utilisation composite
1	1	.00	.00	.00		
	2	.00	.03	.01		
	3	.05	.01	.01		
	4	.00	.01	.06		
	5	.00	.05	.02		
	6	.01	.09	.07		
	7	.06	.05	.01		
	8	.02	.01	.01		
	9	.01	.07	.03		
	10	.51	.29	.47		
	11	.33	.39	.31		
2	1	.00	.00	.00	.00	.00
	2	.00	.03	.01	.00	.00
	3	.05	.01	.00	.00	.00
	4	.00	.01	.06	.00	.00
	5	.00	.05	.02	.00	.00
	6	.01	.09	.07	.00	.00

Collinearity Diagnostics[a]

Model	Dimension	Variance ... Operational BI utilisation composite
1	1	
	2	
	3	
	4	
	5	
	6	
	7	
	8	
	9	
	10	
	11	
2	1	.00
	2	.00
	3	.00
	4	.00
	5	.00
	6	.00

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions		
				(Constant)	Company size 50-99 (ref: <50)	Company size 100-249 (ref: <50)
	7	.774	2.898	.00	.00	.01
	8	.593	3.310	.00	.10	.18
	9	.284	4.782	.00	.74	.58
	10	.080	9.036	.00	.00	.03
	11	.055	10.832	.00	.00	.03
	12	.016	20.127	.46	.00	.00
	13	.011	23.857	.53	.04	.02
	14	.009	27.288	.00	.01	.00

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions: Job: Senior Manager (ref: CEO)	Job: Middle Manager (ref: CEO)	Job: Operational Manager (ref: CEO)	Job: Other (ref: CEO)	Industry: Retail (ref: Manufacturing)
	7	.00	.07	.09	.00	.02
	8	.01	.02	.10	.04	.00
	9	.01	.02	.00	.00	.00
	10	.10	.15	.14	.07	.30
	11	.49	.55	.54	.78	.32
	12	.03	.02	.02	.01	.10
	13	.02	.04	.00	.04	.14
	14	.02	.02	.04	.03	.00

Collinearity Diagnostics[a]

Model	Dimension	Variance Proportions: Industry: Service (ref: Manufacturing)	Industry: Technology (ref: Manufacturing)	Industry: Other (ref: Manufacturing)	Strategic BI utilisation composite	Tactical BI utilisation composite
	7	.06	.05	.01	.00	.00
	8	.02	.01	.01	.00	.00
	9	.00	.04	.02	.00	.00
	10	.29	.39	.27	.03	.02
	11	.36	.17	.33	.00	.00
	12	.14	.07	.11	.34	.04
	13	.06	.09	.09	.19	.01
	14	.00	.00	.00	.43	.93

Collinearity Diagnostics[a]

Model	Dimension	Variance ... Operational BI utilisation composite
	7	.00
	8	.00
	9	.00
	10	.03
	11	.00
	12	.14
	13	.65
	14	.18

a. Dependent Variable: Decision-making quality composite

```
* -------------------- 12. SENSITIVITY: BI-NAMED DMQ ITEM REMOVED ------------
--------

* Following v29's described approach: remove the most directly BI-anchored DMQ
 item
* (BI_Improves_Decisions, which most explicitly states that BI improves decisi
ons)
* and recompute the DMQ composite from the remaining three items.
* This bounds the definitional component of the BI-DMQ association.

COMPUTE DMQ_3item = MEAN(BI_Speed_Accuracy, BI_Data_Inform, BI_Alternative_Sol
utions).
EXECUTE.
VARIABLE LABELS DMQ_3item 'DMQ composite (3 items, BI_Improves_Decisions remov
ed)'.

REGRESSION
  /MISSING LISTWISE
  /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI(95)
  /CRITERIA=PIN(.05) POUT(.10)
```

```
/NOORIGIN
/DEPENDENT DMQ_3item
/METHOD=ENTER Strategic_Mean Tactical_Mean Operational_Mean
/RESIDUALS DURBIN.
```

Regression

Notes

Output Created		18-JUN-2026 22:32:48
Comments		
Input	Data	C:\Users\Hoda\OneDrive\Desktop\files\BI_DMQ_data_v30.sav
	Active Dataset	RawData
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	186
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics are based on cases with no missing values for any variable used.
Syntax		REGRESSION /MISSING LISTWISE /STATISTICS COEFF OUTS R ANOVA CHANGE COLLIN TOL CI(95) /CRITERIA=PIN(.05) POUT(.10) /NOORIGIN /DEPENDENT DMQ_3item /METHOD=ENTER Strategic_Mean Tactical_Mean Operational_Mean...
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Notes

	Memory Required	2740 bytes
	Additional Memory Required for Residual Plots	0 bytes

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Operational BI utilisation composite, Strategic BI utilisation composite, Tactical BI utilisation composite[b]	.	Enter

a. Dependent Variable: DMQ composite (3 items, BI_Improves_Decisions removed)

b. All requested variables entered.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.746[a]	.557	.550	.52144	.557	76.270	3	182	.000

Model Summary[b]

Model	Durbin-Watson
1	2.165

a. Predictors: (Constant), Operational BI utilisation composite, Strategic BI utilisation composite, Tactical BI utilisation composite

b. Dependent Variable: DMQ composite (3 items, BI_Improves_Decisions removed)

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	62.213	3	20.738	76.270	.000[b]
	Residual	49.486	182	.272		
	Total	111.699	185			

a. Dependent Variable: DMQ composite (3 items, BI_Improves_Decisions removed)

b. Predictors: (Constant), Operational BI utilisation composite, Strategic BI utilisation composite, Tactical BI utilisation composite

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.489	.224		2.187	.030
	Strategic BI utilisation composite	.233	.078	.232	2.973	.003
	Tactical BI utilisation composite	.281	.091	.256	3.095	.002
	Operational BI utilisation composite	.361	.075	.345	4.792	.000

Coefficients[a]

Model		95.0% Confidence Interval for B		Collinearity Statistics	
		Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	.048	.930		
	Strategic BI utilisation composite	.078	.387	.401	2.496
	Tactical BI utilisation composite	.102	.461	.357	2.804
	Operational BI utilisation composite	.212	.510	.468	2.135

a. Dependent Variable: DMQ composite (3 items, BI_Improves_Decisions removed)

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions (Constant)	Strategic BI utilisation composite	Tactical BI utilisation composite
1	1	3.956	1.000	.00	.00	.00
	2	.023	13.189	.93	.13	.03
	3	.013	17.355	.04	.41	.01
	4	.008	21.602	.02	.45	.96

Collinearity Diagnostics[a]

Model	Dimension	Variance ... Operational BI utilisation composite
1	1	.00
	2	.03
	3	.83
	4	.14

a. Dependent Variable: DMQ composite (3 items, BI_Improves_Decisions removed)

Residuals Statistics[a]

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	1.5623	4.8654	3.8065	.57990	186
Residual	-1.65684	1.72521	.00000	.51719	186
Std. Predicted Value	-3.870	1.826	.000	1.000	186
Std. Residual	-3.177	3.309	.000	.992	186

a. Dependent Variable: DMQ composite (3 items, BI_Improves_Decisions removed)

```
* -------------------- 13. SAVE PROCESSED FILE --------------------

SAVE OUTFILE='BI_DMQ_data_v30_processed.sav'
  /COMPRESSED.

>Error # 5332 in column 14.  Text: BI_DMQ_data_v30_processed.sav
```

```
>The specified file or directory is read-only and cannot be written to.  The
>file will not be saved.  Save the file with another name or to a different
>location or change the access permissions first.
>Execution of this command stops.

* -------------------- END OF SYNTAX --------------------.
```